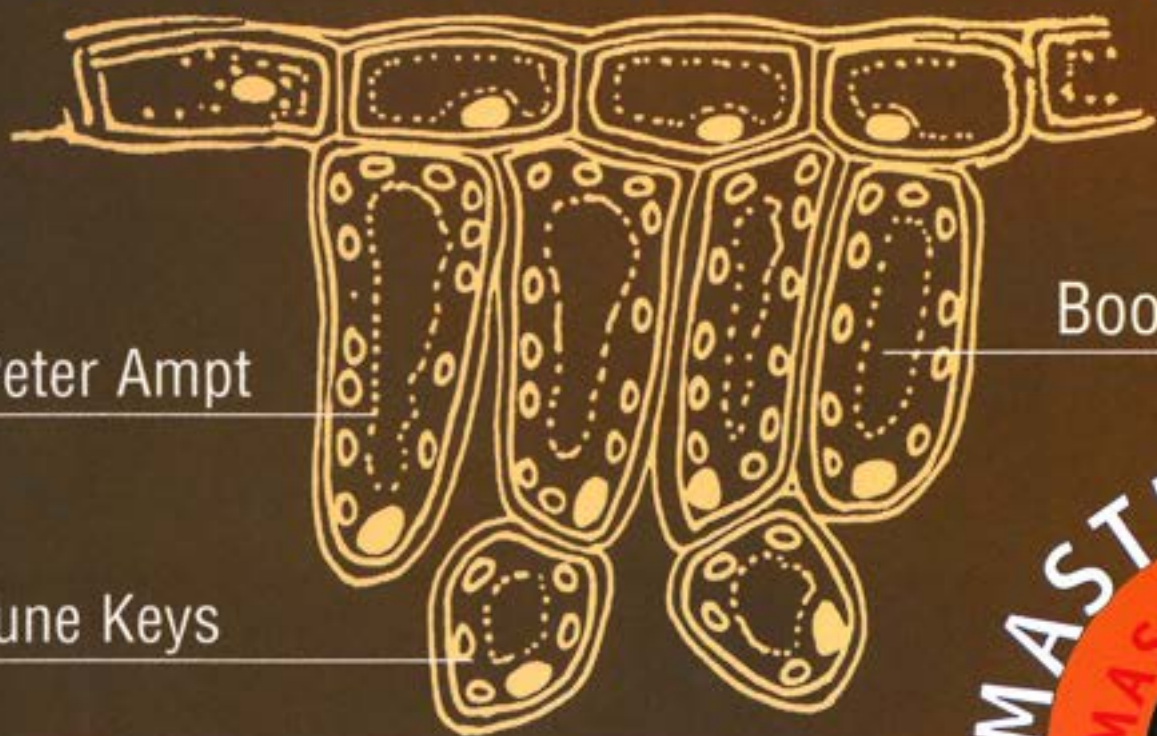


INTERPRETING DATA IN BIOLOGY



Peter Ampt

Book 1

June Keys



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INTERPRETING DATA IN BIOLOGY

Book 1

Peter Ampt and June Keys



Emerald City Books

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Preface

The purpose of this series of books is to provide opportunities for senior secondary students to develop skills in manipulating, reading, describing, analysing and interpreting biological data. In many cases the data is from actual historical experiments from scientific literature and in other cases it has been generated or approximated from real data.

Teachers will find the activities in this book useful as supplementary material for practical work, as materials for relief teachers or as homework. Suggested answers for all questions are provided.

This book contains 17 stand-alone activities. Every activity should take approximately 40 minutes to one hour to complete. Each activity requires students to:

- 1** Read the data, which is presented as tables, graphs, flow charts, diagrams or text.
- 2** Translate or manipulate the data into a form that promotes understanding.
- 3** Describe the trends in the data using a scaffold example as a guide.
- 4** Analyse or interpret the data and relate it to existing theoretical knowledge.

As students become familiar with the format, they will become more able to complete the activities with minimal assistance. Less-able students will usually be able to complete each activity up to Step 3. Many students may require mediation by the teacher to complete the final stage of each activity.

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A food web of a Murray River ecosystem

Introduction

A food web is a diagram that represents the feeding relationships between organisms within an ecosystem. Since food contains energy, food webs also show the movement of energy within an ecosystem. In a food web, the arrows mean "is eaten by" and show the direction that energy passes within the ecosystem. Food webs can be drawn from data collected in the field. This data may be in the form of direct observations of one organism feeding on another or by the analysis of the gut contents of consumers.

The population of each species in an ecosystem is influenced by the amount of food available and by the number of predators it has. If the amount of food increases, then the population will increase. This may, in turn, lead to an increase in the number of predators. By using a food web, it is possible to predict how a change in population of one species may affect other species.

Aim

To draw a food web using data collected in the field, and to use this food web to predict the effects of changes in the ecosystem.

Data

Name of organism	Picture of organism	Observations
tadpole		seen feeding on algae in the river
water snail		seen feeding on algae in the river
mosquito larva		seen feeding on algae in the river
dragonfly nymph		seen feeding on mosquito larvae
Long-necked tortoise		gut contents include partially digested mosquito larvae, tadpoles, water-snails and dragonfly nymphs plus intestinal worms were observed living in the gut
Murray Cod		gut contents include partially digested dragonfly nymphs and Long-necked tortoises
White-necked heron		seen feeding on Long-necked tortoises and small Murray Cod
Tiger snake		occasionally seen feeding on Long-necked tortoises
goanna		occasionally seen feeding on Long-necked tortoises
fox		occasionally seen feeding on Long-necked tortoises

Morgan, D. (ed), *Biological Science: the web of life*, Australian Academy of Science, Canberra, 1988.

Interpreting the information

1 Complete the table below

Type of organism	Explanation	Examples in the Murray River ecosystem
producer (plants)	produce food from sunlight	
1st order consumers	eat producers	
2nd order consumers	eat first order consumers	
3rd order consumers	eat second order consumers	
4th order consumers	eat third order consumers	
top consumers	not eaten by any other organism	

2 Draw a food web for the Murray River ecosystem.

Note: When drawing a food web, it is conventional to put the producers (green plants) at the bottom of the diagram, place first order consumers (herbivores) above the producers, then place second order consumers (carnivores) above first order consumers and so on until all organisms have been included. Arrows are then added from the producers to the first order consumers, then from the first order consumers to the second order consumers and so on until all of the feeding relationships have been shown.

- 3
 - a Where does the energy in this food web come from? _____
 - b Through which organism does this energy enter the food web? _____
 - c What biological process is responsible for the energy entering the food web? _____
- 4 Which activity causes energy to flow through the food web? _____
- 5
 - a Name a top consumer in the food web. _____
 - b Through which organisms has energy flowed to feed this top consumer? _____
- 6 Parasites are organisms which feed from another living thing (a host) without killing it. From the food web above, name a parasite and its host. _____
- 7 Name an organism which is both a second and a third order consumer. _____
- 8 Pollution of the river resulted in a dramatic decline in the population of the mosquito larvae and tadpoles. Explain how this will affect the populations of:
 - a dragonfly nymphs _____
 - b Long-necked tortoises _____
 - c water snails _____
 - d Murray Cod _____
- 9 Name some organisms that are involved in recycling that have not been included in this ecosystem. _____

Food web data — how do we know who eats what?

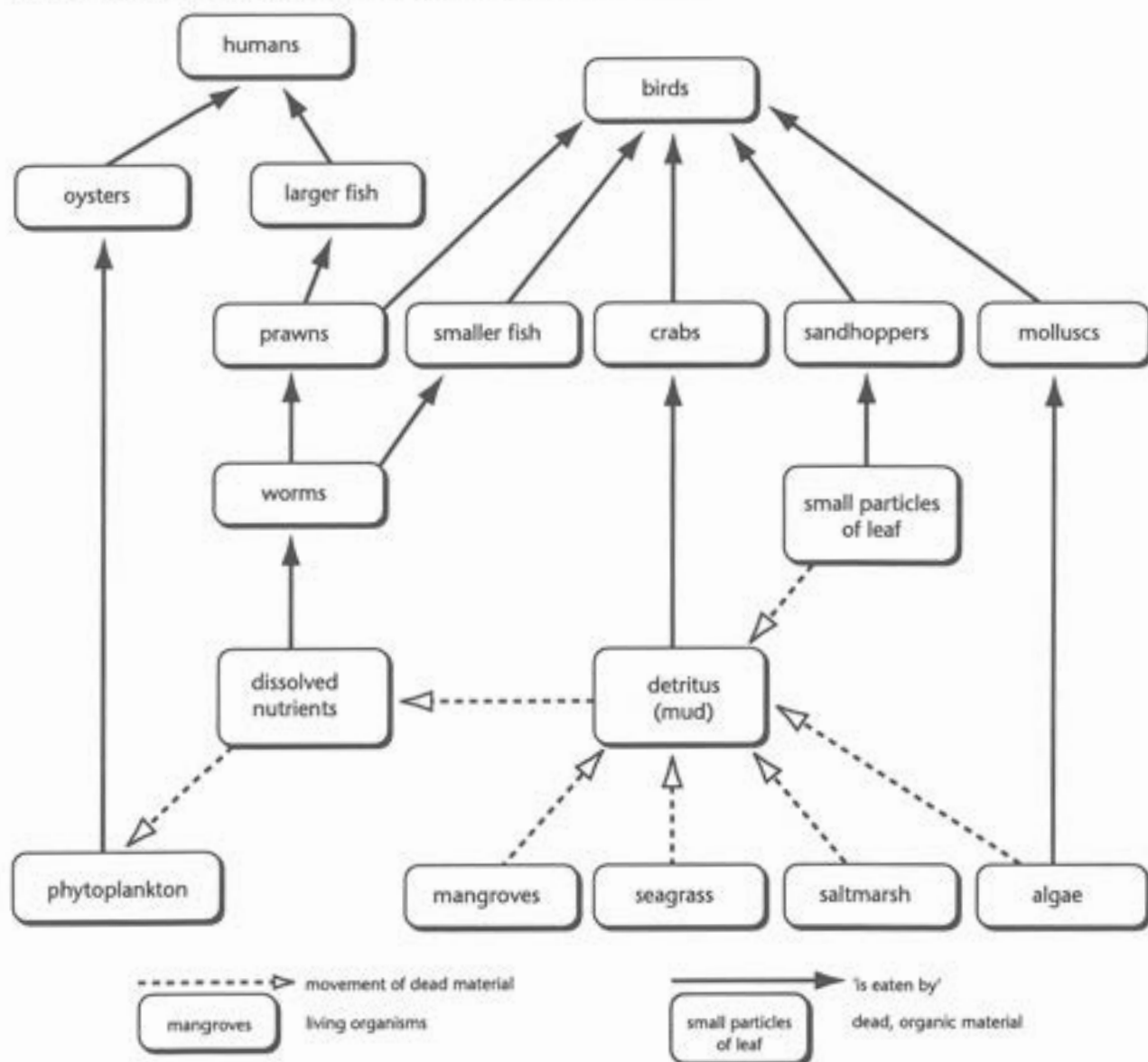
Introduction

Wetlands on the coast support a wide variety of life. They also provide most of the dissolved nutrients that are the basis of life in the oceans. Scientists find out 'who eats what' in a complex ecosystem such as a wetland by observing the behaviour of organisms and examining their gut contents to find out what they have eaten. As scientists find more information, they draw food webs like the one below to show feeding relationships and the flow of energy in ecosystems. Food webs can be used to predict the likely effects of changes to ecosystems.

Many coastal wetlands have been converted into canal developments. Developers dredge shallow waters to make deeper canals and build up swampy areas to provide suburbs where all houses have water frontages.

Aim

To use a food web to predict the likely effects of canal developments on ecosystems.



Underwood, A.J. & Hutchings, P., *Australia's seashores*, Collins/Australian Museum, Sydney, 1987.

Interpreting the data

- 1 Why are some organisms called producers? Identify the organisms that don't eat other organisms, then colour these organisms green. You can tell they don't eat anything because there are no unbroken arrows going to them from other organisms.
- 2 What happens to mangrove, seagrass, saltmarsh and algal material when it dies? _____
- 3 Which organisms rely directly on detritus for food? _____
- 4 What is the food source for oysters? _____
- 5 List all the first order consumers (the animals that eat producers and are often called herbivores).

- 6 List 5 types of animals that are eaten by birds.

- 7 List two types of animals that rely on worms for food. _____
- 8 Canal developments remove most of the mangroves, seagrass and saltmarsh. What will be the effect of canal developments on the first order consumers? Give reasons for your answer.

- 9 Do you think oysters would be adversely effected by canal developments? Explain.

- 10 How would canal development impact on the population of birds? Explain.

Environmental effects on the abundance of sand hoppers on sandy beaches

Introduction

There are several species of small amphipod crustaceans or 'sand hoppers' that live on sheltered, sandy beaches around Sydney. They 'surf' up the beach on waves then burrow quickly into the sand, feeding on algae and the remains of dead organisms. They themselves are food for juvenile fish and seabirds. Marine ecologists at the Australian Museum have studied them intensively because amphipods are thought to be more sensitive to oil pollution than other species. As a result, changes in their population may prove to be a useful indicator of the extent of oil pollution and/or ecosystem damage. The data in this activity is a small section of the data from this study.

Aim

To find any links between the abundance of amphipods and the environmental factors of water temperature and wavepower (the force generated by waves on the beach).



An amphipod crustacean

Method

The amphipods were sampled by removing a core of sand 2 cm wide and 5.5 cm deep, dumping it into a fine mesh sieve and counting the animals as they hopped around. Ninety cores were taken every month for two years from one section of Silver Beach near Kurnell, south of Sydney. Environmental data were collected at the same location.

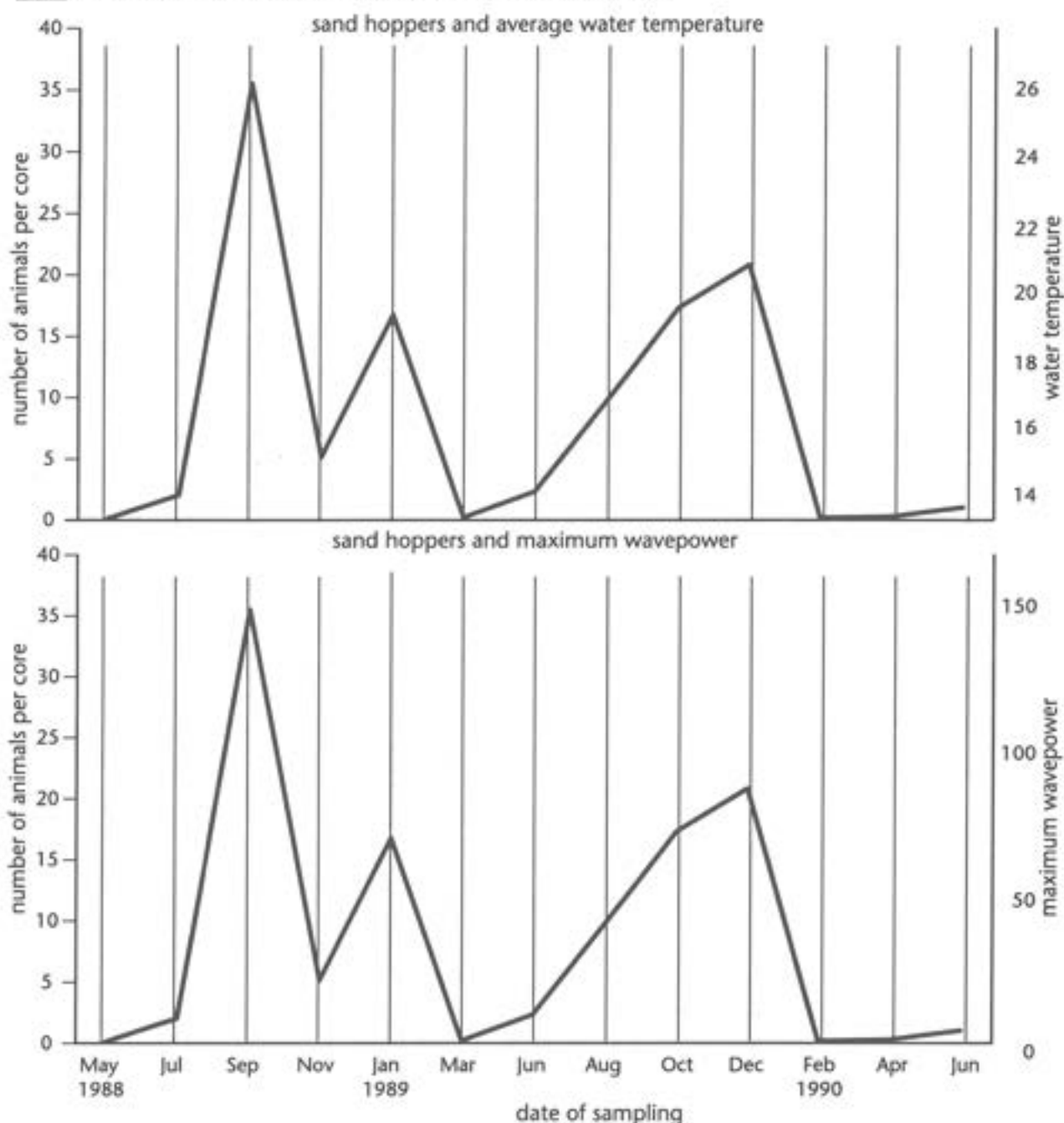
Results

Date of sampling	Average number of animals per core	Water temperature (°C)	Maximum wavepower (N.ms ⁻¹ m ⁻¹ x 10 ⁻³)
May 1988	0.12	18.7	64
June	2.42	16.4	155
July	2.04	18	45
August	7.17	17	59
September	35.51	17	27
October	12.38	21	25
November	5.14	23	22
December	9.20	20.5	39
January 1989	16.74	26.5	21
February	5.01	24.8	22
March	0.21	24.5	37
April	0.09	22.2	29
May	0.34	21	46
June	2.33	14.2	39
July	1.64	16	98
August	9.76	16.2	31
September	20.47	20	18
October	17.32	21	42
November	27.94	20.5	20
December	20.82	23.5	27
January 1990	5.06	21.6	18
February	0.18	23	38
March	0.17	22.5	53
April	0.28	21.3	30
May	0.16	20.8	30
June	1.03	16.5	17

Jones, A.R., Murray, A. & Marsh, R.E., *Hydrobiologia*, vol. 223, pp.119-126, 1991.
Murray, A., Australian Museum, personal communications, 1999.

Activities

- 1 On the first graph, plot the data for water temperature.
- 2 On the second graph, plot the data for maximum wavepower.



- 3 In which season are the animals most abundant and in which season are they least abundant?
- 4 Is there a seasonal pattern for wavepower?
- 5 Describe the seasonal pattern for water temperature.
- 6 Can you describe any links between the average number of animals and either wavepower or water temperature? If so, describe the links.
- 7 Do you think changes in wavepower or water temperature caused the change in the number of animals? Justify your answer.
- 8 What factors other than wavepower and water temperature do you think could cause a change in the average number of animals?

Nest predation in Australian urban environments

Introduction

Many bird species live in urban environments where many people enjoy watching them. Birdwatchers in many urban areas have reported that the number of small birds has been decreasing. They have also reported that the number of currawongs (a predatory, native bird) have been increasing. Even though it seemed that currawongs were causing the decline in the number of small birds, there were other possible explanations such as other birds, cats, kookaburras or possums.

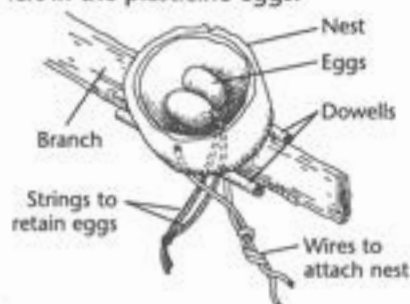
In 1993, ecologists from the Australian Museum carried out an experiment to test whether currawongs were eating the eggs of the smaller birds and causing their numbers to fall. They distributed 2000 artificial nests containing soft plasticine eggs to volunteers across Australia. The volunteers put the nests into trees in their backyards, and observed any animals visiting the nest for two weeks. At the end of the two week period they took the nest down and sent it, with their observations, back to the Museum. At the Museum, the scientists identified the nest predators from the beak, tooth and claw marks they had left in the plasticine eggs.

Aim

To analyse the data from the experiment to determine whether currawongs are significant predators of small birds in urban environments.

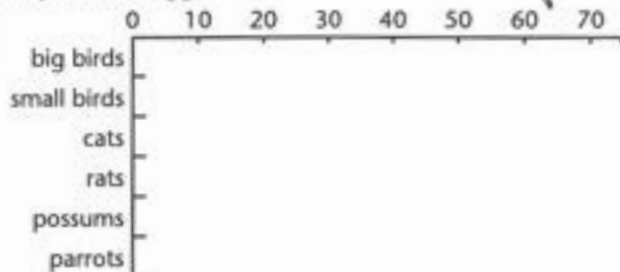
Data

Draw bar graphs that represent the data.



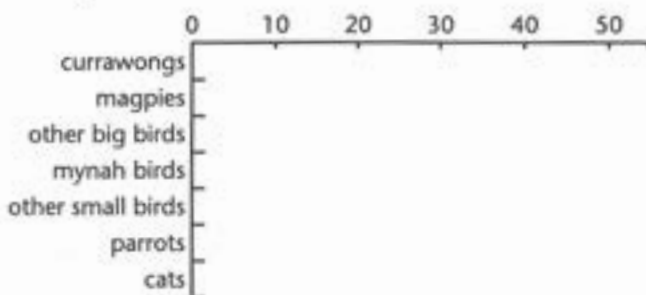
1 Identifying nest predators using markings left in plasticine eggs

Type of predator	% of Imprints in plasticine eggs
big birds	64
small birds	30
cats	3
rats	3
possums	3
parrots	2



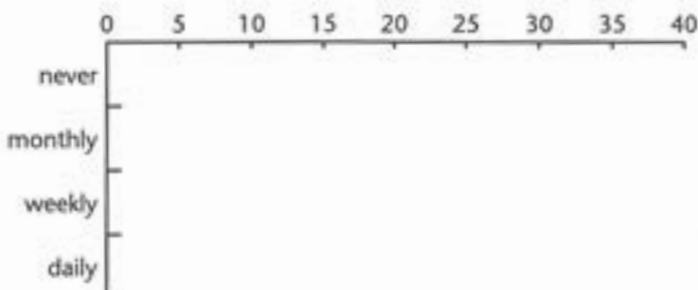
2 Identifying nest predators using direct observation by volunteers

Type of predator	% of witnessed predation
currawongs	52
magpies	13
other big birds	4
mynah birds	10
other small birds	17
parrots	2
cats	3



3 Predation rate of eggs in areas where currawongs are seen

How often do you see currawongs?	% predation
never	18
monthly	25
weekly	32
daily	40



Major, R.E., Gowing, G. & Kendal, C.E. *Australian Journal of Ecology*, vol 21, pp.399-409, 1996.

Interpreting the data

- 1 Describe the evidence from the data that supports the hypothesis that currawongs are the major predators of small birds in urban environments. _____
- 2 Describe the evidence from the data that suggests cats have little effect on the population of small birds. _____
- 3 Cats are believed to find nests by watching adult birds feeding their young. What effect does this information have on the evidence you described in question 2? _____
- 4 Read each statement below and use the evidence provided by the data above to decide which box or boxes most accurately describes each statement. The first one is done for you.

Statement	definitely true	might be true	no evidence	might be false	definitely false
Big birds caused 64% of the imprints on the plasticine eggs.	✓				
13% of the predation observed by the volunteers was by currawongs.					
There is more predation of nests in gardens where currawongs are often observed.					
Cats are major predators of small birds					
Currawongs eat more small bird eggs than any other predators.					
Predation of small birds by currawongs is greater than predation by cats.					
Currawongs are a significant nest predator in urban environments.					

Recommendations

The following recommendations were made by the ecologists who coordinated the study to increase the numbers of small birds in urban environments. Next to each recommendation, write the evidence from the study that supports the recommendation.

Recommendation	Evidence from the study that supports the recommendation
Plant more dense shrubs in gardens to provide small birds with more protection from predators.	
Reduce the number of berry-producing plants in gardens. These plants attract large birds like currawongs because they provide them with food.	
Do not encourage large birds into gardens by feeding them.	

Further studies

- 5 What are the limitations of this study? _____
- 6 What further studies would you carry out to find out more about the decline of small birds in urban environments? _____

Spontaneous generation, the cell theory and scientific process

Introduction

Until the nineteenth century, most people believed that all species had been spontaneously created from dust, dirt and other non-living sources. This belief is called spontaneous generation. Francesco Redi (1626-1698), an Italian scientist, conducted experiments that challenged this belief. However, despite the thorough and scientific approach taken by Redi, people continued to believe in spontaneous generation for hundreds of years afterwards. Today the prevailing belief is that all living things are made of cells and all cells come from pre-existing cells. This is known as 'The cell theory'.

Kirk, A., *Biology Today*, 2nd edn, CRM Random House, New York, 1975.

Aim

To analyse the work of Redi, identify features of his work that follow scientific process, describe how his work challenged the idea of spontaneous generation and explain why people continued to believe in spontaneous generation.

Francesco Redi made careful observations of the way dead bodies decay. His observations and experiments were as follows:

STAGE 1

He watched dead snakes decay in an open box.

'...I saw that they were covered in worms of a conical shape and apparently without legs...intent on devouring the meat, increasing meanwhile in size, and...likewise increased in number; but, although of the same shape, they differed in size, having been born on different days.'

After they finished devouring the meat, the worms disappeared.

STAGE 2

Redi then repeated the experiment with one minor change: he sealed the box after the worms had appeared so that nothing could escape. He noticed two types of worms that both changed 'from soft to hard, resembling chrysalides (cocoons) formed by caterpillars, silkworms, and similar insects'. He then watched as two types of fly emerged from these egg-shaped objects. He realised that the worms (maggots) were the immature forms of the flies and 'began to believe that all worms found in meat were derived directly from the droppings of the flies, and not from the putrefaction of the meat'.

STAGE 3

Redi set up a new experiment to test his belief. He set up four wide-mouthed flasks, put a different type of meat into each flask and sealed them. He then set up an identical set of four flasks but left them open. He observed that the meat in the open flasks quickly became covered with flies but that no flies appeared in the sealed flasks, even though the meat in them decayed. He repeated this experiment many times and reached the same conclusion every time: the worms did not appear as a result of spontaneous generation but as a result of adult flies making contact with the meat.

STAGE 4

However, he realised that when he sealed the jars, he also excluded air from them and that it could be a lack of fresh air that prevented the maggots from growing. So he set up another experiment in which flies were prevented from entering the flasks by a fine mesh which would allow the movement of air. Again he found no worms on the meat unless flies had been able to enter the jars.

Questions

- 1** Draw lines to link each stage with the process or processes that Redi used. Two are done for you.

make careful observations

Stage 1

design an experiment

formulate a hypothesis

Stage 2

vary the one factor that is relevant to the hypothesis

modify your hypothesis

Stage 3

use a control

Stage 4

replicate your experiment

- 2** Write at least one valid conclusion for each stage of Redi's work. The first one is done for you.

Stage 1 — Worms appear in dead animals, devour the meat and then disappear.

Stage 2 —

Stage 3 —

Stage 4 —

- 3** Through his experiments, what was Redi able to show that challenged the idea of spontaneous generation?

- 4** Despite Redi's work, why do you think people continued to believe in spontaneous generation?

Size of cells and cell organelles

Introduction

The cell theory states that all organisms are made of cells and that all cells come from pre-existing cells. But what do cells look like and how big are they? And does the size of cells vary in different organisms and in different parts of the same organism?

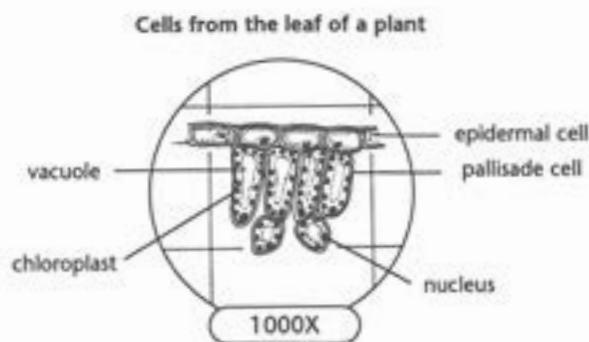
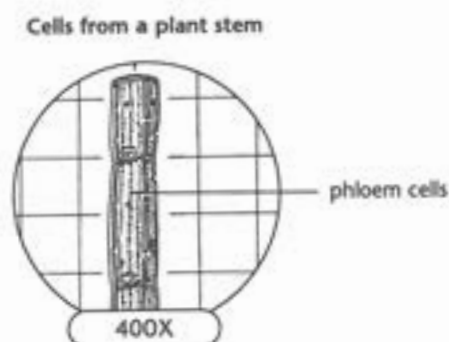
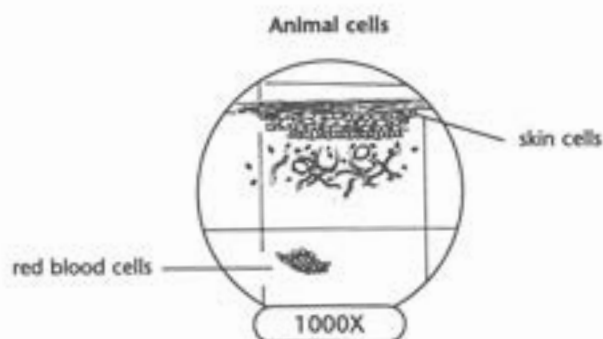
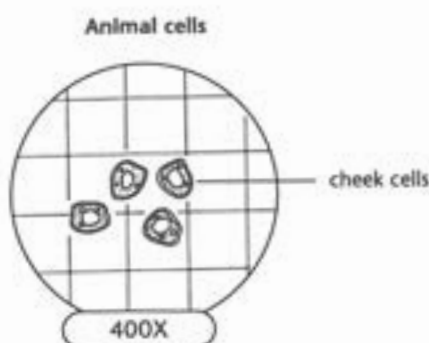
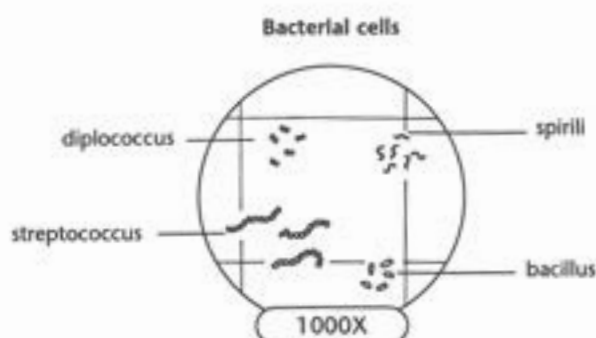
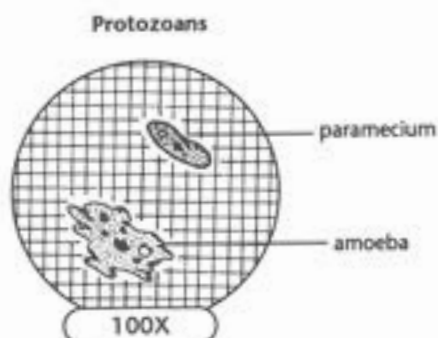
Aim

Firstly, to estimate the size of cells using diagrams drawn from several fields of view and secondly, to use that information to write a report about the size of cells and cell organelles.

Data

Look at the diagrams of cells drawn in the fields of view below. The lines in the grids are 100 μm apart.

Note: 1 μm is one millionth of a metre.



What to do

- 1** Use the grids to estimate the length and width of each cell. The lines on each grid are 100 μm apart. Record your answers in the table below. (Remember, 1 μm is one millionth of a metre.)

Plant cells	size (μm)	Animal cells	size (μm)	other cells	size (μm)	plant cell organelles	size (μm)
phloem		cheek		bacterial		nucleus	
palisade		skin		amoeba		vacuole	
epidermal		red blood		paramecium		chloroplasts	

Interpreting the data

- 2** Which cells are the longest?
3 Which cells are the smallest?
4 Describe the variation in the size of plant cells.
5 Describe the variation in the size of animal cells.
6 How do the size of bacterial cells compare to the size of animal cells?

Writing a report

Reports classify and describe phenomena in general and/or specific terms. There are usually two stages in a report: the classification and the description.

Classification - Classifies the phenomena being described, locates it in time or space and previews the rest of the description to follow.

Description - Each feature of the phenomena is described in a separate paragraph.

- 7** Complete the description stage of this report about cell size.

Classification stage

All living things are made of tiny parts called cells. All cells are very small but some cells are much smaller than others. Even though cells are very small, they contain organelles which are even smaller.

Description stage

Plant cells are ...

Animal cells are ...

Other cells such as bacteria, fungi and protozoa ...

Cell organelles also vary in size. In plant cells, the biggest organelles are ...

Questions requiring research or interpretation

- 8** Why are all cells microscopic in size?
9 How do the size of bacterial cells compare to the size of viruses? (This question requires further research).

Light and photosynthesis

Introduction

A scientist called Engelmann did a series of ingenious experiments in the early part of the twentieth century to investigate the effect of light on photosynthesis. At the time he did these experiments, Engelmann was aware that light, water and carbon dioxide were necessary for plant growth and that plants produced starch and oxygen. However, he did not know which chemicals or structures in the plant carried out photosynthesis or whether the oxygen produced came from water or carbon dioxide. The accepted formula for photosynthesis at the time was:

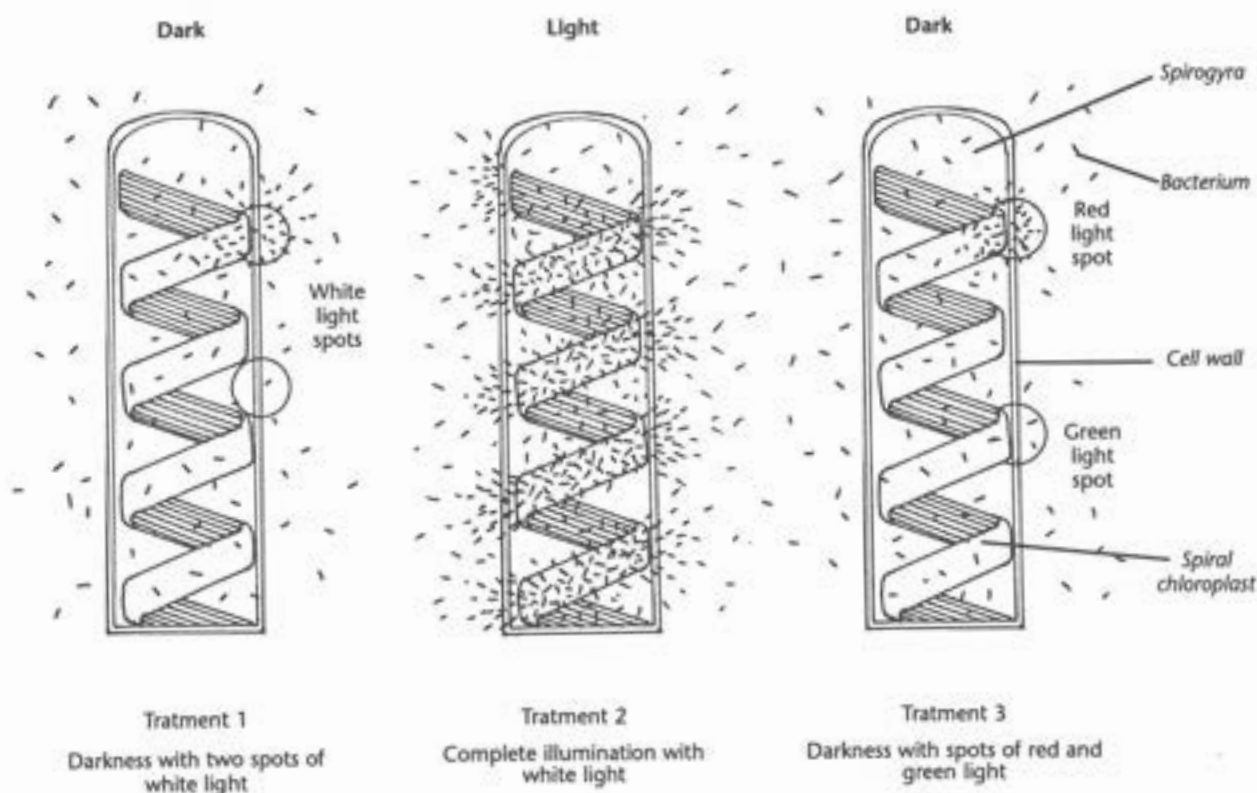


In one of his experiments, Engelmann used *Spirogyra*: a type of alga that has relatively large cells containing spiral-shaped chloroplasts that are visible under a light microscope. He put the alga in water with some aerobic bacteria that grow rapidly where oxygen is abundant. Then he shone some light on different parts of the spiral chloroplast and observed the growth of the bacteria.

Hall, D.O. & Rao, K.K., *Photosynthesis* 5th ed, Cambridge University Press, London, 1995.

Aim

To investigate the results of Engelmann's experiments and relate them to our current knowledge of photosynthesis.



Analysing the experiment

- 1 What was the purpose of using *Spirogyra* in the experiment? _____
- 2 Why did he use aerobic bacteria? _____
- 3 What do you think were the aims of the experiment? _____

Observations

- 4 In treatment 1, what light conditions were necessary to cause bacterial growth? _____
- 5 In treatment 3, what light conditions were necessary to cause bacterial growth? _____
- 6 In treatment 2, where on the spiral chloroplast did the bacteria grow most rapidly? _____
- 7 Write a general statement about light and the growth of aerobic bacteria. _____

Making links

- 8 What does the growth of bacteria in treatment 1 indicate about photosynthesis? _____
- 9 What does the growth of bacteria in treatment 3 indicate about photosynthesis? _____
- 10 Why do you think treatment 2 was included in the experiment? _____

Conclusions

- 11 What do you think were the aims of this experiment? _____
- 12 Write two valid conclusions that you could make as a result of this experiment. _____
- 13 As a result of your conclusions to this experiment, what changes could you make to the formula for photosynthesis that was accepted at the time? _____
- 14 Compare the current formula for photosynthesis with the formula that was accepted in the early part of the twentieth century. What aspects of photosynthesis were still unknown? _____
- 15 How could you improve the experimental method used by Engelmann? _____
- 16 Why is it significant that photosynthesis does not occur in green light? _____

Using radioactive carbon to trace the path of carbon in plants

Introduction

Carbon enters a plant as carbon dioxide and is used during photosynthesis. Carbon is then moved throughout the plant to where it is needed. In 1945, two scientists, Rabideau and Burr, used radioactive carbon 13 (^{13}C) to investigate how carbon moves through a plant. They suspected that phloem tissue was used to transport carbon compounds. To test this hypothesis, they supplied carbon dioxide that contained ^{13}C to French bean plants. Some time later they were able to detect the presence of radiation and thus ^{13}C at various points on the plant. They removed a ring of tissue from around the stem of some of the bean plants and then measured the amount of ^{13}C in different parts of the plants. Their data is summarised in the diagram below.

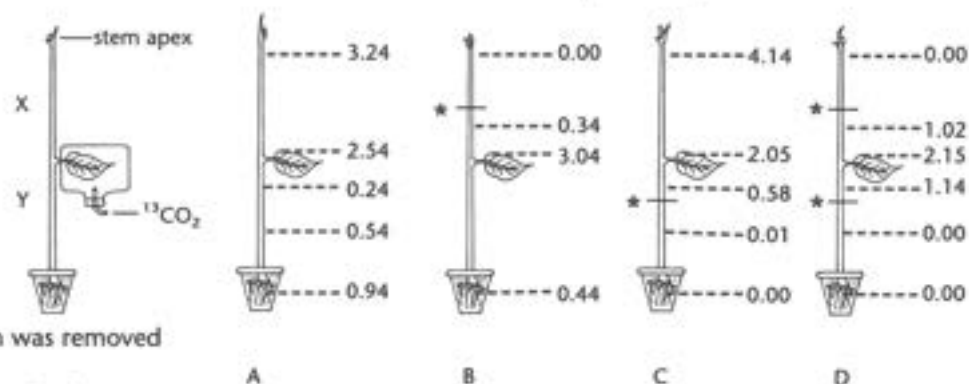
Ridge, I. (ed) *Plant Physiology*, Hodder and Stoughton, London, 1991.

Aim

To use Rabideau and Burr's data to describe the movement of carbon compounds in plants.

Data

The numbers indicate the amount of ^{13}C in different parts of the plants.



* places where phloem was removed

Describing the data

- 1 Did ^{13}C move out of the same leaf to which it was supplied?
- 2 Where did the ^{13}C go?
- 3 What effect did cutting the phloem (when the ring of tissue was removed from the stem) have on the movement of ^{13}C ?
- 4 Which part of plant A had the largest amount of ^{13}C ?

Making links

- 5 In plant A, did ^{13}C move up or down?
- 6 In plant B, did ^{13}C move up or down?
- 7 In plant C, did ^{13}C move up or down?
- 8 In plant D, did ^{13}C move up or down?

Interpretation

- 9 Why was there no ^{13}C at the stem apex of plants B and D?
- 10 Why was there no ^{13}C in the roots of plants C and D?
- 11 Does ^{13}C move in the phloem? Explain your answer using evidence from the data.
- 12 What substance in the plant contains ^{13}C ?
- 13 Does the glucose formed by photosynthesis move up or down the plant?

What happens to the air that we breathe in?

Introduction

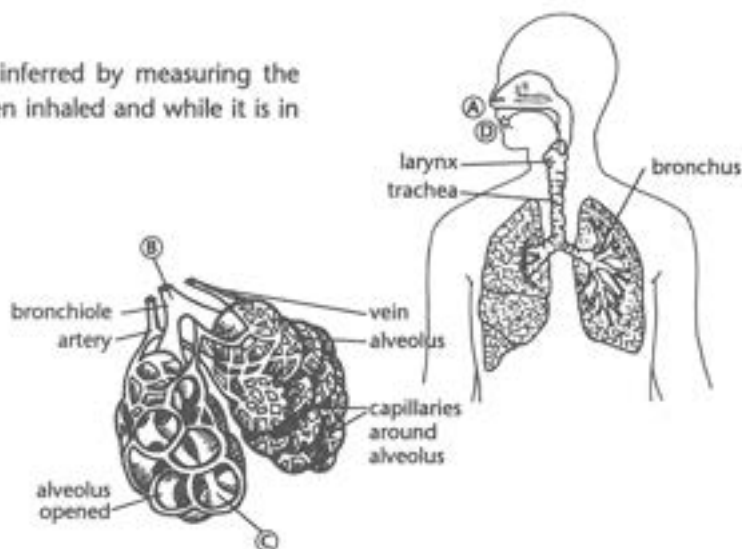
The function of lungs in mammals can be inferred by measuring the composition of air before and after it has been inhaled and while it is in the lungs.

Aim

To determine what happens to inhaled air when it is inside a mammal.

Data

Percentages of gases in air before and after it has been inhaled and while it is in the lungs.

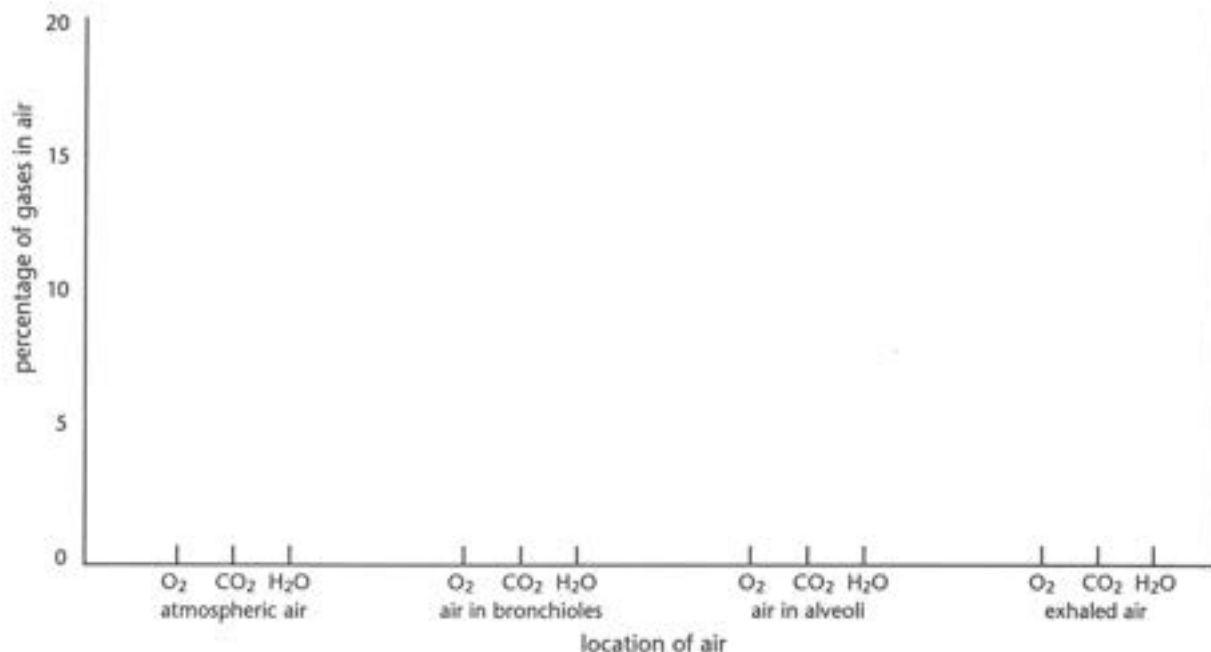


Gas in air (Percentage)	Ⓐ Air before entering mouth and nose	Ⓑ Air in bronchiole before entering alveoli	Ⓒ Air in alveoli	Ⓓ Air exhaled from mouth and nose
nitrogen N_2	78.62	74.09	74.90	74.50
oxygen O_2	20.84	19.67	13.60	15.70
carbon dioxide CO_2	0.04	0.04	5.30	3.60
water vapour H_2O	0.50	6.20	6.20	6.20

Guyton, A.C., *Textbook of Medical Physiology*, W B Saunders Co, Philadelphia, 1976.

Processing data

- Graph the percentages of oxygen, carbon dioxide and water vapour in each location.



Interpreting the data

- 2 Describe what happens to the percentage of water vapour in the air during breathing. _____

- 3 What is the possible cause (or causes) for the change in the percentage of water vapour in the air during breathing? _____

- 4 During breathing, what effect or effects does the change in the percentage of water vapour have on the body? _____

- 5 Describe what happens to the percentage of oxygen in the air during breathing. _____

- 6 What is the possible cause (or causes) for the change in the percentage of oxygen in the air during breathing? _____

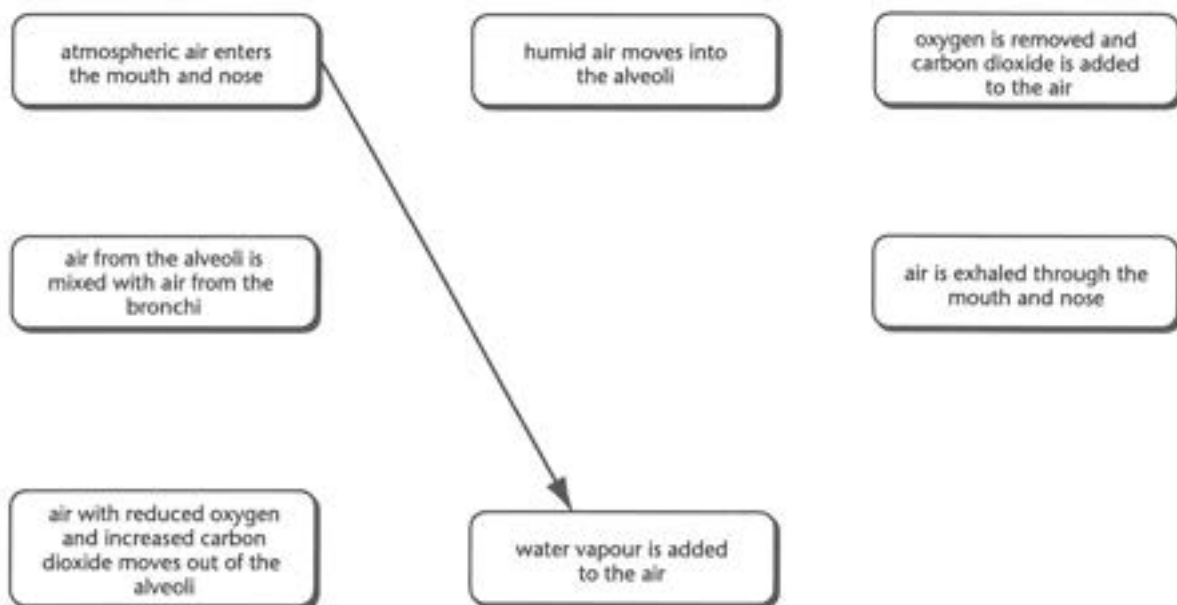
- 7 During breathing, what effect or effects does the change in the percentage of oxygen have on the body? _____

- 8 Describe what happens to the percentage of carbon dioxide in the air during breathing. _____

- 9 What is the possible cause (or causes) for the change in the percentage of carbon dioxide in the air during breathing? _____

- 10 What effect or effects does the change in the percentage of carbon dioxide have on the body? _____

- 11 What happens to air during breathing? Using the data, draw arrows on the diagram below. The first one is done for you.



Gas exchange in amphibians

Introduction

Gas exchange in most organisms involves the absorption of oxygen and the release of carbon dioxide. Gas exchange is necessary because these organisms undergo aerobic respiration. All animals have moist surfaces (called respiratory surfaces) through which they can absorb oxygen and release carbon dioxide in a process called 'gaseous exchange'. Respiratory surfaces in multicellular organisms are often in or on specialised organs such as lungs and gills. Adult amphibians (ectothermic animals) have two organs for gas exchange: lungs and a moist skin.

Aim

To investigate the role of lungs and skin in oxygen absorption in amphibians.

Data

TABLE 1

Oxygen uptake by frogs (millilitres of O_2 per kg bodyweight per hour).

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
skin	55	55	54	52	50	49	49	51	58	60	58	56
lungs	84	77	68	60	42	29	20	27	62	112	112	95

TABLE 2

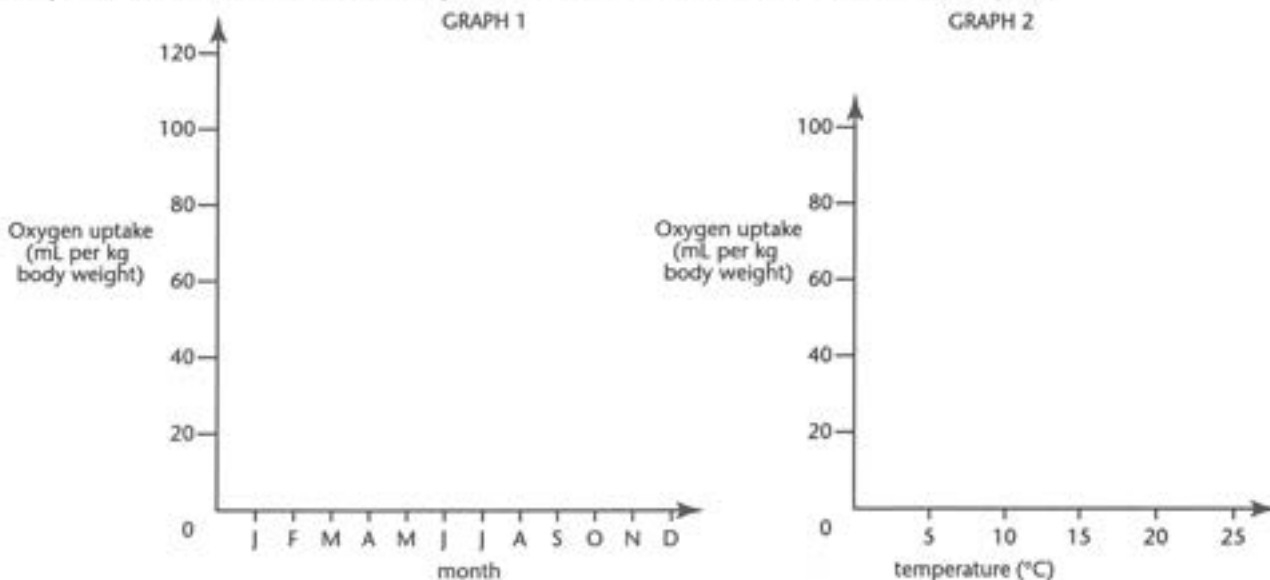
Oxygen uptake by the lungs and skin of toads in different temperatures (millilitres of gas per kg bodyweight per hour).

Temperature	oxygen in lungs	oxygen in skin
5°C	8	16
15°C	65	44
25°C	81	43

Schmidt-Nielsen, K., *Animal Physiology: Adaptation and environment*, Cambridge University Press, Cambridge, 1991.

Graphing the data

Graph the data from Table 1 onto Graph 1, below, and the data from Table 2 onto Graph 2.



Describing the data

- 1 Describe how oxygen uptake by the skin changes throughout the year. _____

- 2 During which months is oxygen uptake by the lungs greater than oxygen uptake by the skin? _____

- 3 During which months is overall oxygen uptake lowest? _____

- 4 During which seasons is
 - a oxygen uptake highest? _____
 - b oxygen uptake lowest? _____
- 5 At low temperatures, which organ is more important for oxygen uptake? _____

- 6 At high temperatures, which organ is more important for oxygen uptake? _____

Making links

- 7 During which season would you expect amphibians to have a higher respiratory rate? Why? _____

- 8 Explain why oxygen uptake is highest in the season you chose in question 4 a above. _____

- 9 Why does oxygen uptake by the skin remain relatively constant all year round? _____

- 10 How can amphibians greatly increase the amount of oxygen absorbed by the lungs at certain times of the year? _____

- 11 Contrast the oxygen demands of endotherms (animals that maintain their own body temperature) throughout the year with the oxygen demands of ectotherms (animals that have the same body temperature as their surroundings) throughout the year. _____

Comparing the digestive systems of different mammals

Introduction

The function of the digestive system is to break down food into particles that are small enough to be absorbed into the bloodstream. The nature of the digestive system of a mammal is determined by its diet. Foods such as hard leaves or fibrous grasses are difficult to break down. These foods can only provide nutrition if they stay in the digestive system long enough to be fermented by the microorganisms that live in the digestive system. By contrast, meat and fleshy fruits and vegetables are generally easier to digest and pass through the digestive systems of the animals that eat them more quickly.

Aim

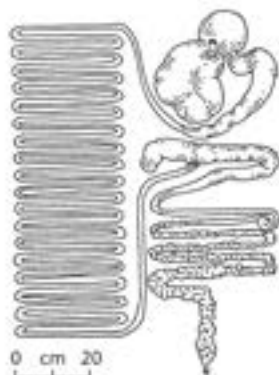
To compare the digestive systems of Koalas, humans and sheep and relate them to the diet of each mammal.

Table: Organs of digestion and their functions

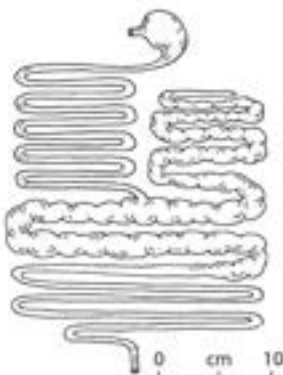
Organ	Description	What happens there?
oesophagus	tube	Δ transfers food from mouth to stomach
stomach	muscular bag with opening at each end	Δ food is mixed with digestive juices Δ digestion of protein usually begins
enlarged or additional stomach	chamber or enlarged tube in some animals	Δ microbiological fermentation Δ nutrients absorbed into the blood
small intestine	long thin tube	Δ digestion of fats and carbohydrates Δ nutrients absorbed into the blood
caecum (see-cum)	tubular extension of the large intestine, enlarged in some animals	Δ microbiological fermentation Δ nutrients absorbed into the blood
large intestine	thick tube which is enlarged in some animals	Δ water in food is absorbed into the blood Δ microbiological fermentation Δ nutrients absorbed into the blood

Data

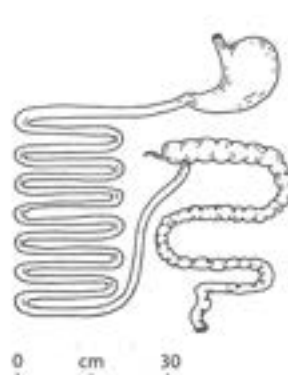
Sheep
Body Length: 110 cm



Koala
Body Length: 51 cm



Adult Human
Body Length: 180 cm



Stevens, C.E. & Hume, I.D., *Comparative physiology of the vertebrate digestive system*, Cambridge University Press, Cambridge, 1995.

Animal	Diet
Koala	eucalyptus leaves
human	a wide range of plants, animals and animal products
sheep	fibrous plants

Making measurements and tabulating data

Use the diagrams on the previous page to estimate the length of each digestive system. Complete the table below.

Animal	total length of digestive system (cm)	length of body (cm)	length of digestive system divided by length of body	Does the animal have enlarged organs for microbial fermentation?
koala				very large elongated caecum
human				no
sheep				large complex stomach and enlarged caecum

Making links

- Is there a link between the length of the digestive system and the length of the body? _____
- Why is it helpful to calculate the length of an animal's digestive system divided by its body length? _____
- Is there a link between the diet and the length of an animal's digestive system divided by its body length? Describe the link. _____
- Which animal has the shortest digestive system compared to its body length? Explain how this relates to its diet. _____
- Which animal has the longest digestive system compared to its body length? Explain how this relates to its diet. _____
- Compare the digestive systems of the sheep and the Koala. What are their similarities and differences? _____
- Compare the diets of sheep and Koalas. Can you explain the differences in the length of their digestive systems in terms of their diets? _____
- What does the Koala have that compensates for its relatively short digestive system? _____

Making conclusions

- What conclusions can you draw about diet and digestive systems from this data? _____

Chemical evolution and the origin of life

Introduction

Most biologists believe that the first life evolved on Earth more than 3 billion years ago, about 1 billion years after the Earth was formed. Biologists have hypothesised that before life began, a process of chemical evolution occurred in which the chemical building blocks of life were formed throughout the following series of events:

The inorganic (non-living) constituents of living things were present on the Earth when life began.

Environmental conditions that were favourable for chemical reactions existed. These chemical reactions formed more complex chemicals from the inorganic constituents.

These complex chemicals became the building blocks of life and included amino acids (which form proteins), sugars and nitrogen bases (which form nucleotides then combine to form nucleic acids), sugars (which combine to form polysaccharides), glycerin and fatty acids (which combine to form fats and other lipids) and ATP (Adenosine triphosphate — an energy transport molecule).

These building blocks became arranged into membrane systems which formed the first cells. These cells were prokaryotic bacterial cells and they did not need oxygen. (There was no free oxygen in the atmosphere at this time in Earth's history). Cells that do not require oxygen are called anaerobic cells.

There is a large amount of evidence that has given scientists confidence that chemical evolution did happen and that it made the evolution of the first living organisms and biological evolution possible.

Aim

To investigate evidence that chemical evolution occurred and that it led to the origin of life.

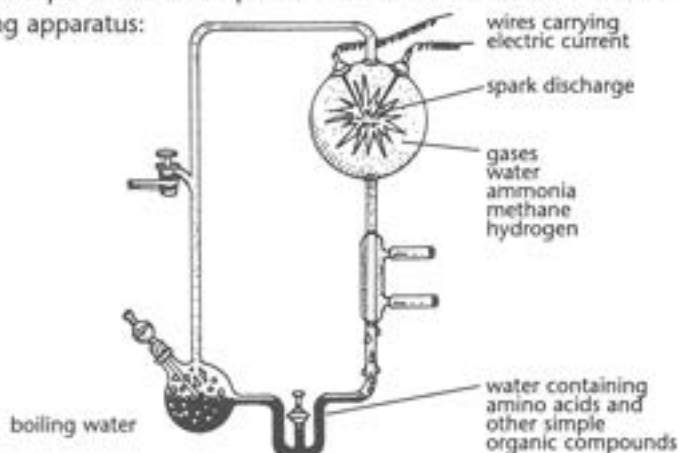
Data

EVIDENCE BOX 1

Geological evidence suggests that hydrogen (H_2), methane (CH_4), ammonia (NH_3), nitrogen (N_2) hydrogen sulfide (H_2S) and water (H_2O) were present in the atmosphere more than 3 billion years ago. In addition, rain could have leached elements (such as sulfur) and minerals from rocks into the sea. Evidence also suggests that high levels of lightning, ultra violet radiation and radioactive decay were present on Earth more than 3 billion years ago.

EVIDENCE BOX 2

In 1952, Miller conducted an experiment which provided evidence that chemical evolution may have taken place. He set up the following apparatus:



After one week, a variety of amino acids and other organic compounds were present.

EVIDENCE BOX 3

Sidney Fox has shown that amino acids can combine in water solutions to form polypeptide chains which become more complex when subject to dry heat. As a result, proteins can form. When placed in hot water solutions, these proteins form microspheres consisting of a double-layered protein wall. Also, these proteins occasionally produce bud-like growths which form into new microspheres.

In addition, Aleksandr Oparin demonstrated that nucleic acids and polysaccharides will also form stable droplets in water solutions.

Modern viruses are similar to aggregates of nucleic acid that were probably present in the early stages of chemical evolution. Viruses are able to penetrate host cells and become part of this cell.

EVIDENCE BOX 4

Many bacteria living today require anaerobic (oxygen-free) conditions. They obtain energy from chemical reactions using chemicals that were present on Earth before life began. For example:

Chlorobium uses hydrogen sulfate in photosynthesis and produces sulfur;

Thiobacillus uses sulfur and produces sulfate ions;

Desulfovibrio uses sulfate ions and produces hydrogen sulfide;

Pseudomonas uses carbon dioxide and hydrogen and produces carbohydrates.

EVIDENCE BOX 5

Fossil evidence of the existence of bacteria-like organisms (between 3.1 billion years and 800 million years old) has been found in South Africa, California and Australia.

Kirk, D., *Biology Today*, 2nd ed, CRM Random House, New York, 1975.

Purves, W.K. & Orions, G.H., *Life: the Science of Biology*, Sinaur Assoc. Inc, USA, 1987.

Curtis, H. & Barnes, N., *Biology*, Worth Publishing, USA, 1989.

Describing the data

FROM EVIDENCE BOX 1:

1 What chemicals were present on Earth before life began? _____

2 What conditions existed on Earth before life began? _____

FROM EVIDENCE BOX 2:

3 Which chemicals were used in Miller's experiment? _____

- 4 What conditions were applied in Miller's experiment? _____

- 5 What were the results of Miller's experiment? _____

- 6 What conclusions can be drawn from Miller's work? _____

FROM EVIDENCE BOX 3:

- 7 Which of the points in the introduction is supported by the evidence in this box? _____

FROM EVIDENCE BOXES 4 & 5:

- 8 What support does the fossil evidence of bacteria give to the concept of chemical evolution? _____

Interpretation

- 1 What components of the theory of chemical evolution (in the introduction) are backed by the evidence given here? _____

- 2 What components of the theory of chemical evolution are not backed by the evidence given here? _____

The role of prokaryotes in the evolution of life

Introduction

The name prokaryote means 'before the nucleus'. Prokaryotes (bacteria and cyanobacteria) are single-celled organisms that are among the simplest forms of life. Prokaryotes lack a nucleus and the internal membrane structures of eukaryotes. They were the first living things on Earth and their presence changed the Earth in a very significant way. It was in prokaryotes that chemosynthesis, anaerobic respiration, photosynthesis and aerobic respiration evolved.

Chemosynthesis

sulfur, nitrogen, iron or minerals

methane + water + energy → organic food substances + energy + ATP

Anaerobic respiration

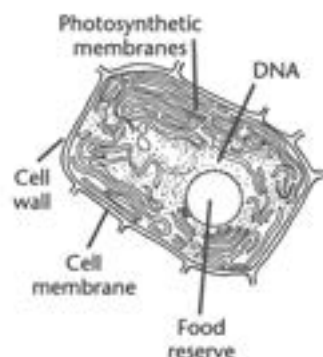
organic food substances + carbon monoxide + ADP → carbon dioxide + water + energy + ATP

Photosynthesis

carbon dioxide + water + light + ADP → glucose + oxygen + ATP

Aerobic respiration

glucose + oxygen + ADP → carbon dioxide + water + energy + ATP



Aim

To use a diagram to describe the role of prokaryotes in the evolution of life on Earth.

Data

Billions of years ago	Environmental conditions	Life forms	Biological molecules
3.5	Atmosphere consists of N_2 , NH_3 , H_2O , CO , CO_2 , CH_4 . Water condenses to form seas. High levels of UV, lightning and radioactive decay.	chemical evolution First simple prokaryotic organisms appear in water, obtaining energy from chemicals and simple anaerobic respiratory pathways that don't involve O_2	First organic molecules appear including amino acids, ATP, RNA, protein and eventually DNA.
3.0			
2.5	O_2 accumulates in the seas and atmosphere, ozone layer forms, protecting land and shallow waters from UV.	Cyanobacteria evolve with photosynthetic pathways that liberate O_2 .	Cytochromes appear making aerobic respiration more efficient.
2.0		Bacteria evolve aerobic respiration which uses O_2 .	
1.5			
1.0		First eukaryotic cells (protozoans and algae) appear with complex respiratory pathways using cyanobacteria as chloroplasts and bacteria as mitochondria.	
0.5		Multicellular organisms evolve: plants from algae and fungi and animals from protozoans.	
present day		Many prokaryotes remain and play an essential role in the recycling of nutrients.	

N_2 = nitrogen, NH_3 = ammonia, H_2O = water, CO = carbon monoxide, CO_2 = carbon dioxide, CH_4 = methane, UV = ultra violet radiation, ATP & ADP = energy carrier, RNA = ribose nucleic acid, DNA = deoxyribose nucleic acid, O_2 = oxygen

Campbell, N.A., *Biology*, Cummings Publishing Co. Inc, USA, 1993.

Purves, W.K. & Orions, G.H., *Life: the Science of Biology*, Sinaur Assoc. Inc, USA, 1987.

Curtis, H. & Barnes, N., *Biology*, Worth Publishing, USA, 1989.

Interpreting the data

- 1 Describe the chemical evolution that had occurred before the first life appeared on Earth? _____
- 2 What type of respiratory pathways did the first prokaryotes use? When the first prokaryotes were on the Earth, why were other pathways, such as aerobic respiration, impossible? _____
- 3 What effects did the evolution of photosynthesis have on the environmental conditions? _____
- 4 Why did the evolution of aerobic respiration occur after the evolution of photosynthesis in cyanobacteria? _____
- 5 Describe how the evolution of eukaryotes depended on prokaryotes being present. _____
- 6 Why do you think some prokaryotes have remained relatively unchanged while others have evolved into eukaryotes? _____

Research task

- 7 Trace the steps of the evolution of higher plants. _____

Making a concept map

Use the data in the table to make a concept map of the role of prokaryotes in the evolution of life. It is best to work with a partner on this activity.

- a Cut out each box.
- b Group related boxes together.
- c Arrange the boxes on a large sheet of paper so that they tell the story of the role of prokaryotes in the evolution of life.
- d Draw arrows between the boxes to link them together.
- e Write labels for the arrows which describe why the boxes are linked together.
- f Compare your concept map with another group's map and discuss any differences.

Cyanobacteria	O ₂ in the atmosphere increases	First prokaryotes appear	N ₂ , NH ₃ , H ₂ O, CO, CO ₂ , CH ₄ in atmosphere	UV, lightning and radiation helped cause chemical evolution
Aerobic respiration	no free O ₂ in the atmosphere	O ₂ in the sea increases	aerobic respiration	anaerobic respiration
chemosynthesis	respiration using no O ₂	obtaining energy from chemicals	eukaryotic organisms appear	protection from UV radiation
respiration using O ₂	first photosynthetic organisms	The first organisms to photosynthesise	cytochromes	use prokaryotic cells as organelles
chemicals that make respiration more efficient	organisms without a nucleus and internal membranes	organisms with a nucleus and internal membranes	cyanobacteria	ozone layer forms

Classification and diversity of plant groups over the past 160 million years

Introduction

The origins of the major groups of plants on Earth today can be determined from fossil evidence. The first plant group was Algae, followed by Bryophytes (mosses and liverworts), Pteridophytes (ferns), Gymnosperms (cone bearing plants) and finally Angiosperms (flowering plants). Table 1 shows the characteristics of the major plant groups. The figure below shows the approximate time that each main group appeared on Earth. Table 2 shows the number of genera in three of these groups since 160 million years ago.

Aim

To graph data on the fossil evidence for the appearance and diversity of the major plant groups and suggest explanations for the changes that have occurred.

Data

TABLE 1: CHARACTERISTICS OF THE MAJOR PLANT GROUPS

Plant group	habitat	true leaves	true roots	xylem & phloem	spores	seeds	flowers	size
Algae	in water	no	no	no	yes	no	no	variable
Bryophytes	in damp, shady areas on land	no	rhyzoids only	no	yes	no	no	5 cm or less
Pteridophytes	in wet areas on land	yes	yes	yes	yes	no	no	up to 4 m
Gymnosperms	on land	yes	yes	yes	no	yes, naked	no	up to 10 m
Angiosperms	on land	yes	yes	yes	no	yes, enclosed	yes	up to 10 m

FIGURE 1: APPEARANCE OF MAJOR PLANT GROUPS

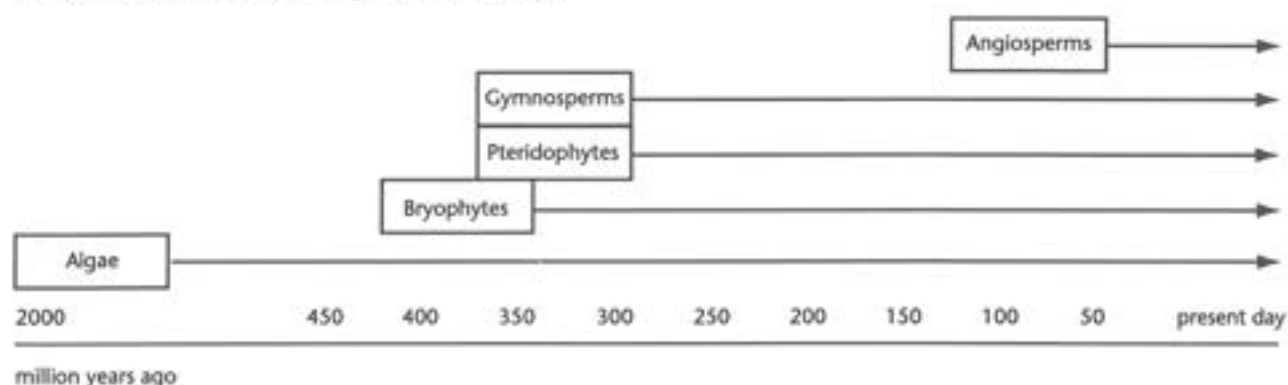


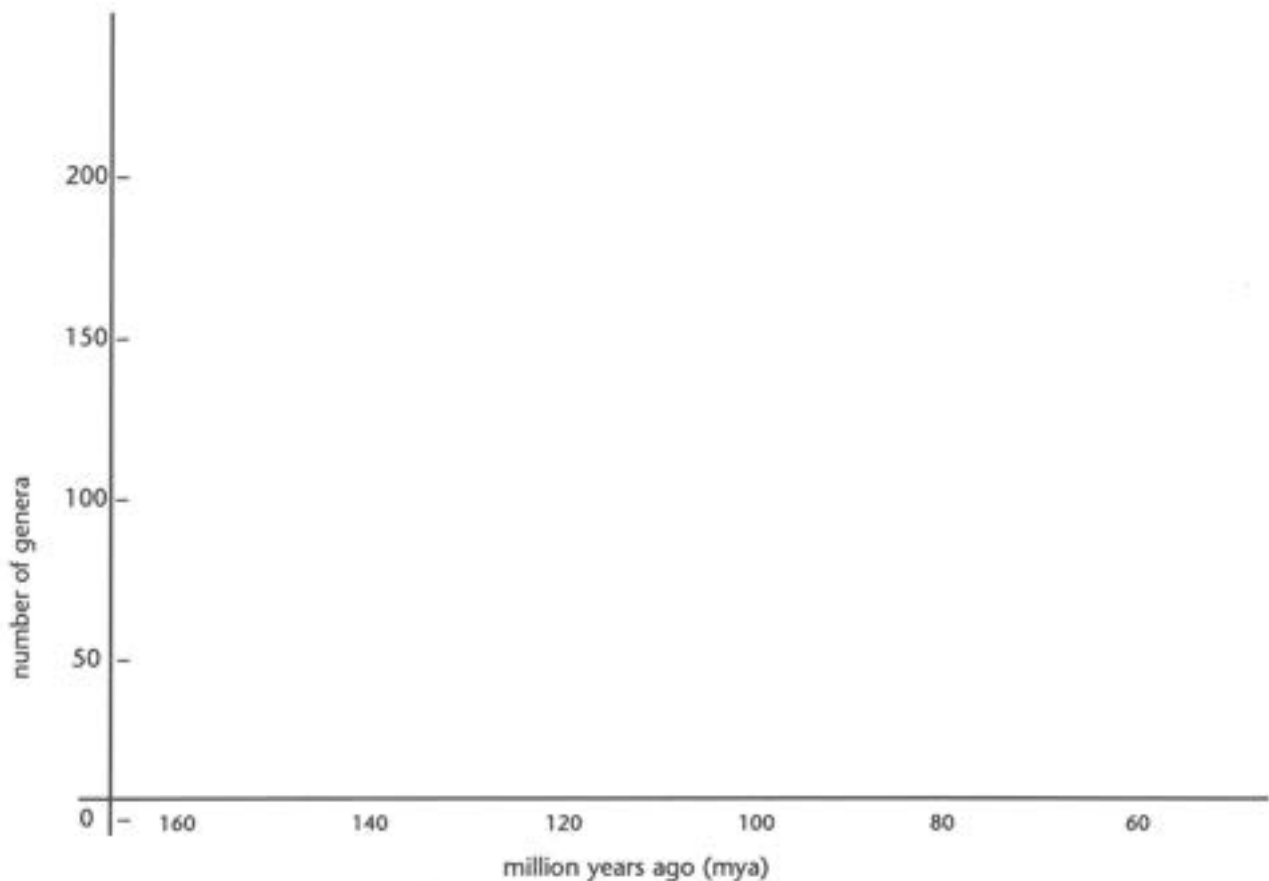
TABLE 2: NUMBER OF GENERA IN THREE PLANT GROUPS FROM 155 TO 60 MILLION YEARS AGO (MYA)

Plant group	155 mya	135 mya	120 mya	105 mya	95 mya	83 mya	78 mya	68 mya	60 mya
Gymnosperms	130	200	202	173	125	116	107	87	107
Pteridophytes	55	100	105	90	75	73	70	60	65
Angiosperms	85	130	145	155	200	210	210	195	245

Cockburn, A., *An introduction to evolutionary biology*, Blackwell Scientific Publications, London, 1991.

Graphing the data

Draw a graph of the number of genera in the above three plant groups over time.



Interpreting the data

Use the graph and the data on the previous page to answer the following questions:

- 1** When did the first plants appear on Earth?
- 2** When did the first land plants appear on Earth?
- 3** What were the first plants to have true roots, true leaves and xylem and phloem?
- 4** What plant group was the last to appear on Earth?
- 5** Describe the trends in the number of genera of Gymnosperms, Pteridophytes and Angiosperms over the past 160 million years.
- 6** Which plant group had the most genera 140 million years ago?
- 7** Which plant group has the most genera at present?
- 8** What characteristics do you think enabled some plants to survive on land? Justify your answer.
- 9** What characteristics may have contributed to the increase in the number of genera of the Gymnosperms until 120 million years ago? Justify your answer.
- 10** What characteristics may have contributed to the increase in the number of genera of the Angiosperms until 120 million years ago? Justify your answer.
- 11** Why do you think the number of genera of Gymnosperms have declined in the past 120 million years? Justify your answer.

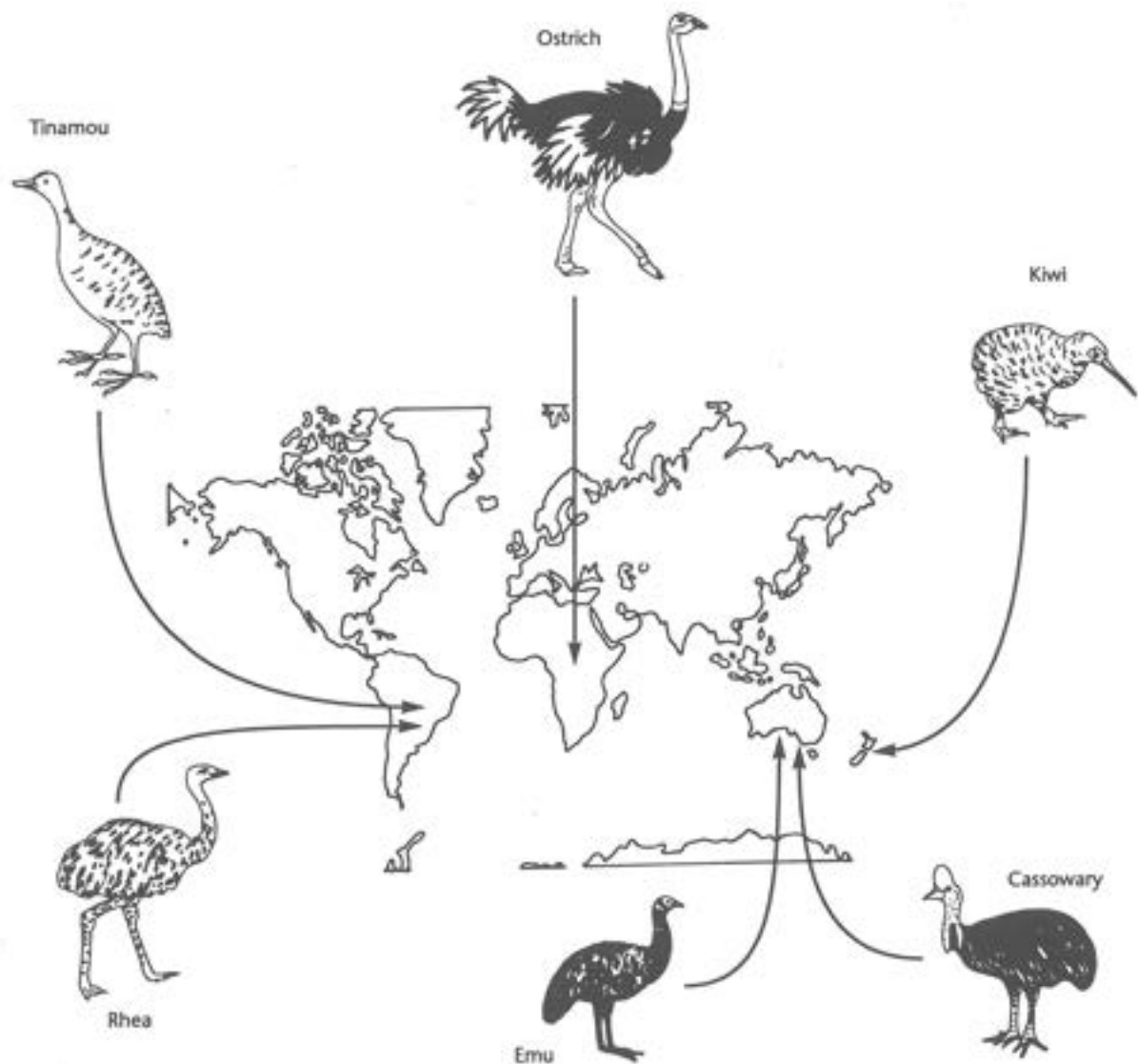
Continental drift and the distribution and evolution of flightless birds

Introduction

There are strong similarities in and significant differences between flightless birds on different continents.

FIGURE 1

Location of flightless birds



Did all these birds evolve from a common ancestor or separate ancestors? Where did their ancestors live? How did these birds end up on three different continents? If they are related, which birds are more closely related? To answer these questions we need evidence. By studying the biochemical similarities (such as DNA sequences) in different species, scientists can show how closely these species are related and estimate how recently they shared a common ancestor. Also, scientists can often draw correlations between these biochemical similarities and continental drift.

Aim

To use present day forms, continental drift and DNA similarities to describe the evolution of flightless birds.

Data

FIGURE 2

Stages in the break-up of Gondwana

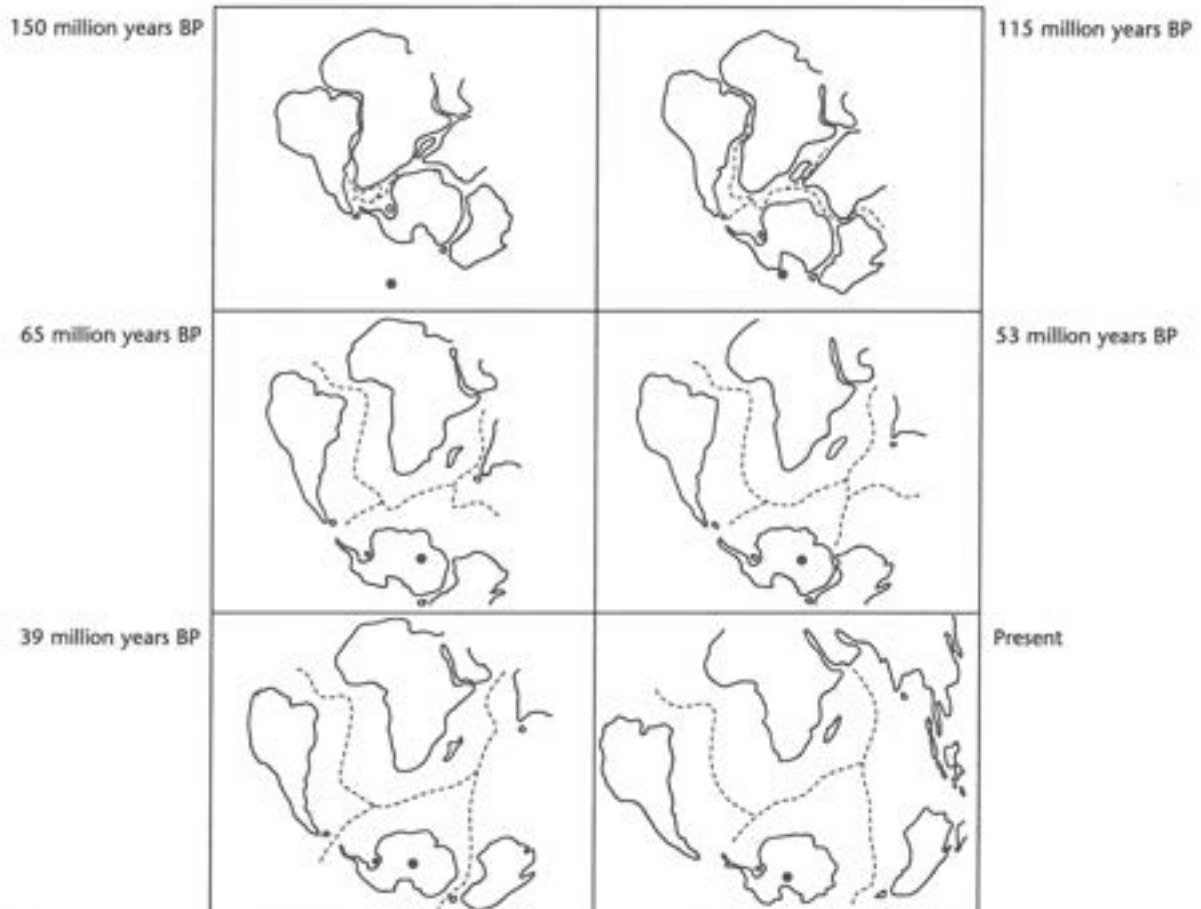
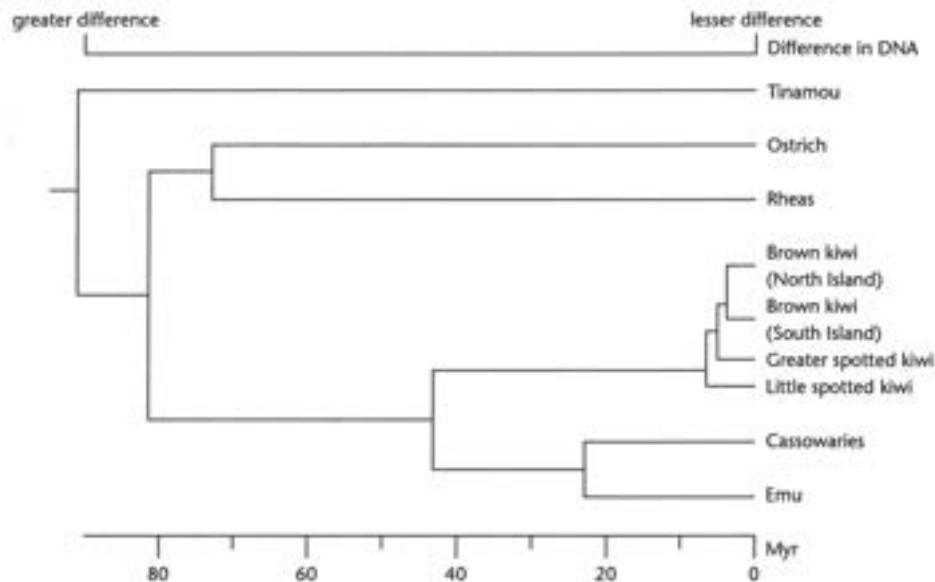


FIGURE 3

Evolutionary links between flightless birds as determined by similarities in their DNA. Birds with a lesser difference in their DNA share a more recent ancestor.



Begon, M., Harper, J.L. & Townsend, C.R. Ecology, Blackwell Scientific Publications, Oxford, 1986.

Describing the data

- 1 Use the preceding data to complete the table below. One is done for you.

Event or evidence	When did it occur? (millions of years ago)
all southern continents were together	
the rift between Australia and the other southern continents	
the opening of the Atlantic Ocean between Africa and South America	
the separation of Australia and New Zealand (not on the diagrams)	40
the presence of a common ancestor of all the present day flightless birds	
the separation of Australian and New Zealand birds from the ostriches and rheas	
the separation of ostriches and rheas	
the separation of kiwis from emus and cassowaries	

- 2 Complete the time line below using information you recorded in the table. One is done for you.

Movements in the continents

millions of years ago

Events in the evolution of flightless birds

	140
	130
	120
	110
	100
	90
	80
	70
	60
	50
Separation of Australia and New Zealand	40
	30
	20
	10
	present day

Making links

- 3 Does DNA information on flightless birds suggest that they all had a common ancestor? Explain. _____
- 4 Does knowledge about continental drift support or refute this evidence? Explain. _____
- 5 Does DNA information about the time ostriches and rheas were separated coincide with the separation of the continents they now live on? Explain. _____
- 6 Does the separation of Australia and New Zealand correspond with the beginning of the isolation of kiwis from emus and cassowaries? Explain. _____

The evolution of Australian plants

Introduction

About 80% of all plant species in the world are endemic to Australia. This means they exist only in Australia. This unique Australian flora is characterised by leathery, spiny and reduced leaves (known as sclerophyll) and can tolerate poor soils and survive frequent fires. It includes many species of Eucalyptus (gum trees), Acacias (wattle and mulga), Casuarinas (she-oaks), hummock grasses, spinifex, grass trees and Banksias. Despite the fact that these species are unique to Australia, the plant families to which these species belong are widespread on other continents.

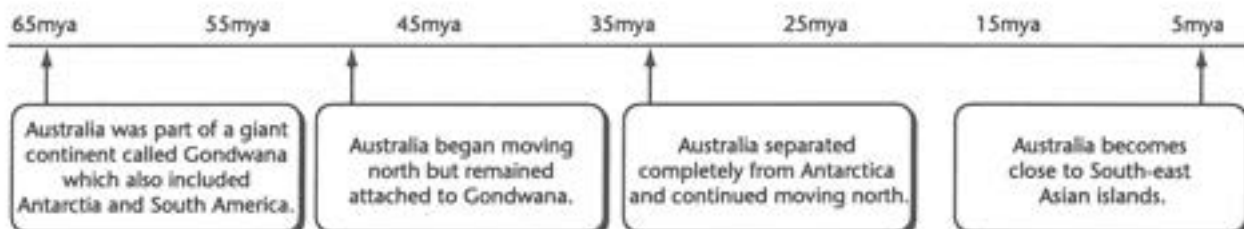
New species evolve if an original population is split and caused to evolve in isolation under different environmental conditions. Australian environmental conditions are low or variable rainfall, poor soils and frequent bushfires.

Aim

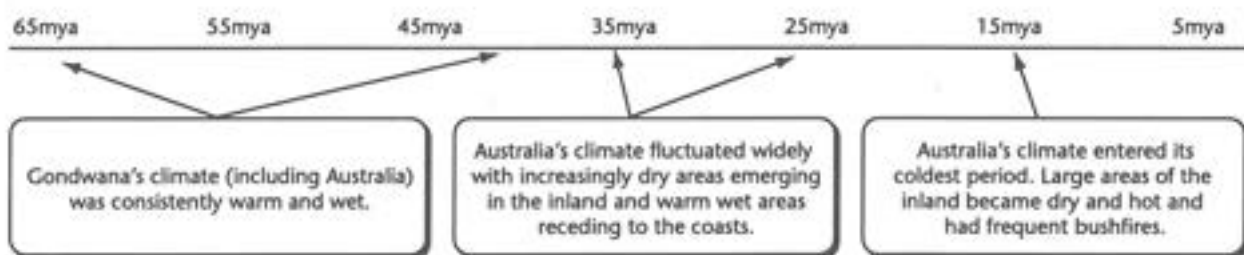
To examine how moving continents, changing climates and the unique features of the Australian environment have caused Australia's unique and distinctive flora to evolve.

Data

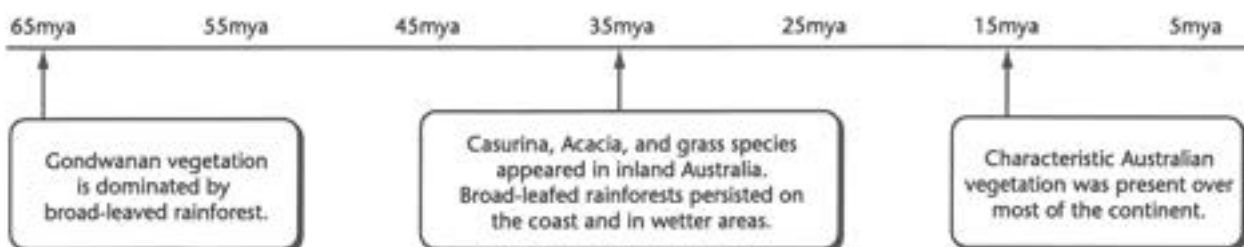
A) THE MOVING CONTINENTS



B) AUSTRALIA'S CLIMATE



C) AUSTRALIA'S VEGETATION



White, M.E., *The Greening of Gondwana*, Reed, NSW, 1986 & 1994.

White, M.E. *After the Greening*, Kangaroo Press, NSW, 1994.

Vickers-Rich, P. & Hewitt-Rich, T. *Wildlife of Gondwana*, Reed, NSW, 1993.

Questions

- 1 Describe Australia's original vegetation. _____

- 2 Where did Australia's original vegetation come from? _____

- 3 Which other areas of the world have vegetation that originated in the same place as Australia? _____

- 4 Under what climatic conditions did Australia's original vegetation evolve? _____

- 5 At what time did Australia's vegetation become isolated from other vegetation? _____

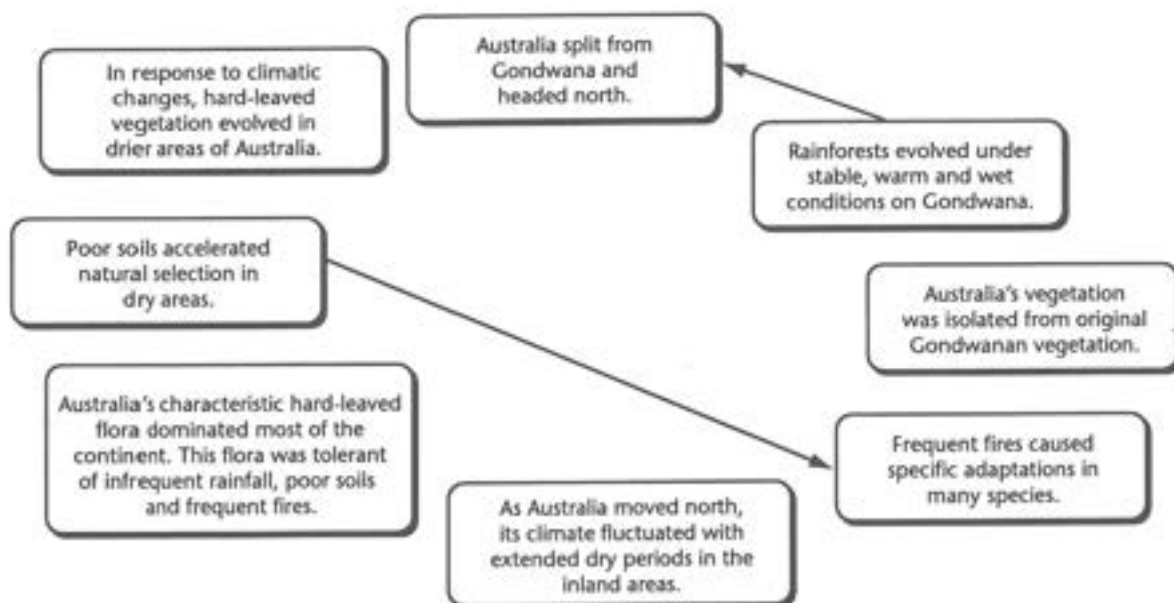
- 6 What changes occurred to Australia's climate after Australia became separated from Gondwana? _____

- 7 How did these climatic changes affect Australia's vegetation? _____

- 8 Why do South American rainforests have some species in common with Australian rainforests? _____

- 9 Explain why Australia's dry-area vegetation is distinctly different to South America's dry-area vegetation despite the fact that both originated in Gondwana. _____

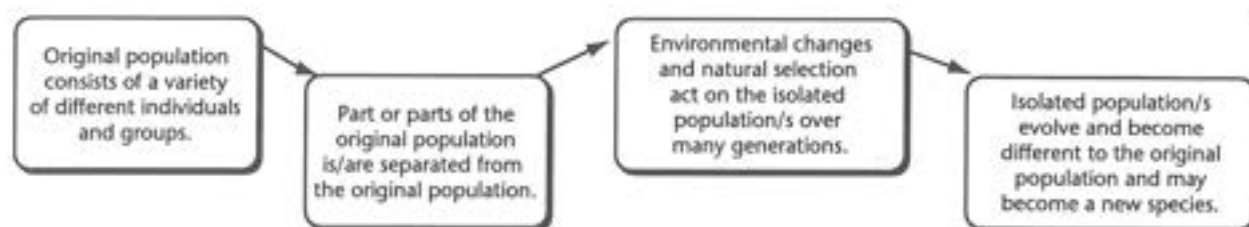
- 10 Join the boxes below with arrows to show the sequence of the events that caused the evolution of Australia's flora. Two are done for you.



The evolution of Australian mammals

Introduction

Why is Australia's mammal fauna dominated by marsupials and monotremes while placental mammals dominate in other parts of the world? Many fossils of mammals that lived during the time when Australia was part of a much larger landmass called Gondwana have been found. Scientists are trying to uncover the story of the evolution of Australia's mammals by studying these and more recent fossils as well as data on geology, climate and vegetation. These data show changes that have resulted in many new species over time. The formation of new species is summarised below.



Aim

To use data about climate, continental drift, vegetation and animal fossils to explain why Australian fauna is dominated by marsupials and monotremes.

Data

Epoch/period and age	World climate and continental drift	Climate and vegetation in Australia	Australian animals
Holocene — from 10 thousand years ago		Similar to today	Similar to today
Pleistocene — from 2 million years ago		Cool/dry climates and low sea levels alternate with warm/wet climates and high sea levels.	Megafauna become extinct. Dingos and people appear.
Pliocene — from 5.2 million years ago	Australia drifts into lower latitudes.	Australia cools and dries out, rainforests decline and eucalypts and grasslands spread.	Modern animals appear, including those specialised in grazing. Many of these become very large and are known as megafauna.
Miocene — from 23.3 million years ago	Australia makes contact with south-eastern Asia.	Lush forests begin to decline and grasslands begin to spread.	Rodents appear.
Oligocene — from 35.4 million years ago	Australia separates from Gondwana.	Increasing dry periods in the interior. Grasslands replace forest communities.	Koalas, kangaroos, possums and other families appear.
Eocene — from 56.5 million years ago	Australia begins moving north but remains connected to Gondwana.	Rainforest covers much of southern Australia and Antarctica.	Australia's oldest marsupials, bats, frogs and snakes appear.
Palaeocene — from 65 million years ago	World climates cool.	Rainforests are spreading across the land.	Mammals diversify on all continents. Marsupials are predominant in South America which is near Australia.
Cretaceous — from 145.6 million years ago	While still part of Gondwana, Australia is located next to Antarctica and South America.	Most of Australia is covered by shallow seas that recede. Flowering plants appear and spread rapidly.	Giant aquatic reptiles appear in shallow seas. Earliest known Australian mammal (platypus-like) and birds appear. Dinosaurs and many other groups become extinct.

continued...

Jurassic — from 210 million years ago		Climate is warm and wet. Conifers, cycads and ferns dominate the land.	Dinosaurs and tiny, mouse-sized mammals appear. First birds appear in the northern hemisphere.
Triassic — from 245 million years ago	Gondwana begins to break up.	Climate becomes warmer and drier.	Mammal-like reptiles are on all continents. Many insects, early amphibians and reptiles appear. No mammals in Australia but present elsewhere.
Before 245 million years ago	Australia is part of Gondwana which is partly covered by glaciers.	As glaciers recede and the climate warms, swamp forests develop.	Insects, fish and early amphibians are plentiful. No reptiles in Australia but present elsewhere.

Archer, M., Hand, S. & Godthelp, H., *Riversleigh: The story of animals in ancient rainforests of inland Australia*, Reed, NSW, 1992.
 Vickers-Rich, P. & Hewitt-Rich, T., *Wildlife of Gondwana*, Reed, NSW, 1993.

Describing the data

- 1 Describe the population of mammals that existed elsewhere in the world before the earliest Australian mammal about 200 million years ago.
- 2 What types of mammals were most common in or around Australia before it started moving north?
- 3 Were the mammals living in Australia at the time it began separating from Gondwana similar to mammals on the other continents?
- 4 What environmental changes took place in Australia after it separated from Gondwana?
- 5 What events coincided with the appearance of rodents in the Miocene Epoch?
- 6 What changes occurred in Australian mammals during the Pliocene Epoch? What were the corresponding changes in environmental conditions during that time?
- 7 What events coincided with the extinction of the megafauna?

Making links

- 8 How might the mammals present in Australia before it split from Gondwana have influenced modern Australian mammals?
- 9 In what ways might the environmental conditions in Australia in the Oligocene Epoch have influenced mammals?
- 10 Suggest two possible explanations for the appearance of rodents during the Miocene Epoch.
- 11 What environmental conditions may have lead to the appearance of the megafauna?
- 12 Suggest some possible explanations for the extinction of the megafauna.

Conclusion

- 13 Why is Australia's mammal fauna dominated by marsupials and monotremes?

Selected answers

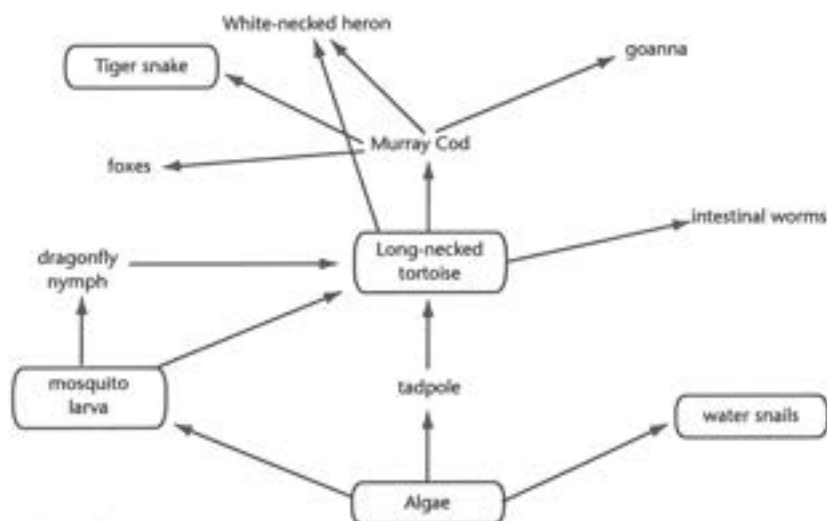
Activity 1: A food web of a Murray River ecosystem

Interpreting the information

1

Type of organism	Explanation	Examples in the Murray River Ecosystem
producer (plants)	produce food from sunlight	algae
1st order consumers	eat producers	tadpole, water snail, mosquito larva
2nd order consumers	eat first order consumers	dragonfly nymph, Long-necked tortoise
3rd order consumers	eat second order consumers	Murray Cod, White-necked heron, Tiger snake, goanna, fox, Long-necked tortoise
4th order consumers	eat third order consumers	White-necked heron
top consumers	not eaten by any other organism	White-necked heron

2

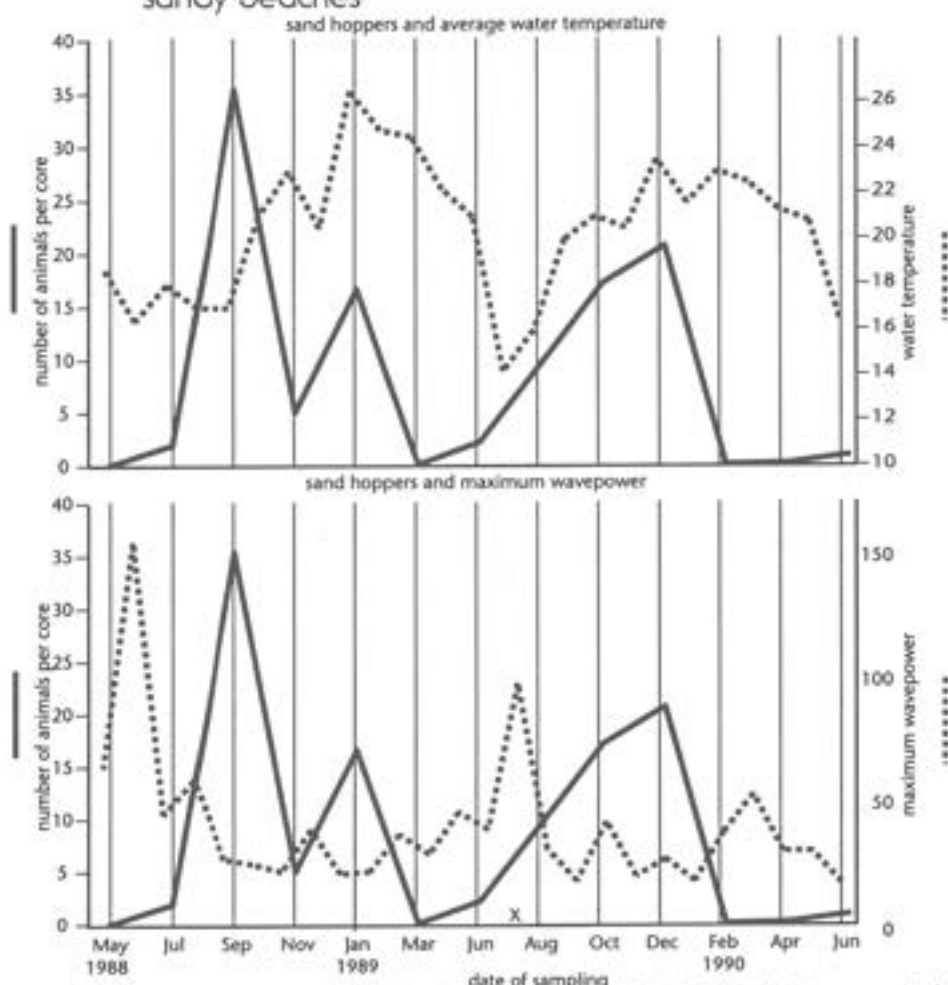


- 3 a From the sun
b Through the algae
c Photosynthesis
- 4 Feeding
- 5 a White-necked heron.
b Algae, tadpole, dragonfly nymph, Murray Cod, White-necked heron.
- 6 Intestinal worms in the Long-necked tortoise.
- 7 Long-necked tortoise.
- 8 a Dragonfly nymph population would decline because they have less food.
b Long-necked tortoise population would decline because they have less food.
c Water snails would increase because there are fewer animals eating algae, their main source of food.
d Murray Cod would decline slightly because they have fewer dragonfly larva to eat but would still have tortoises to eat.
- 9 Decomposers such as bacteria and fungi.

Activity 2: Food web data — how do we know who eats what?

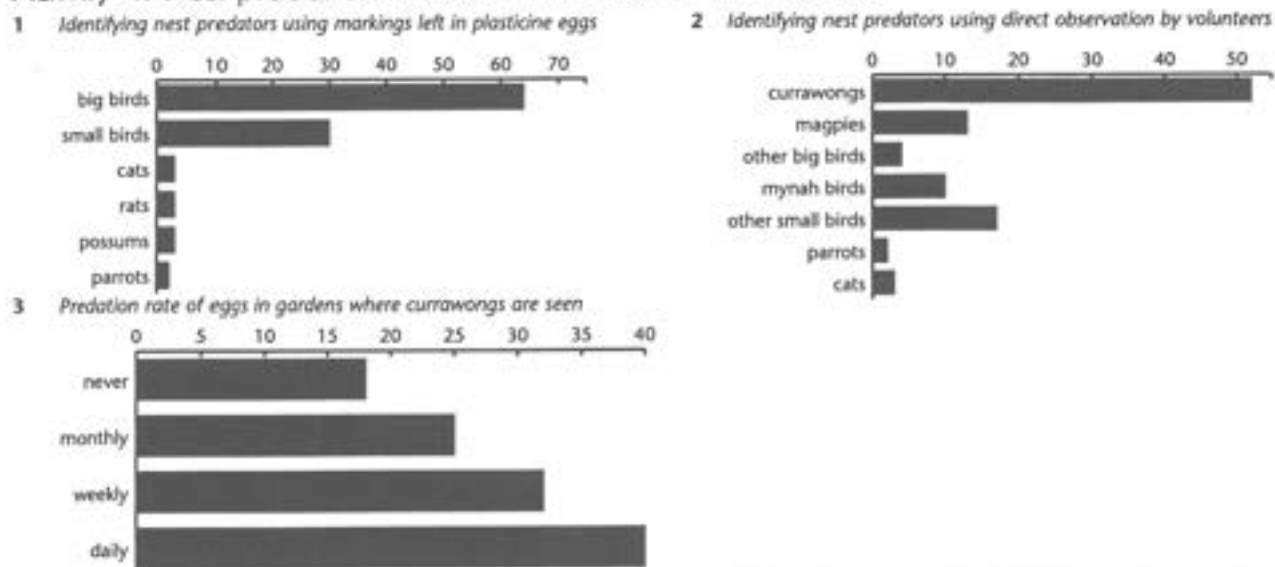
- 1 They are called producers because they use the sun's energy to produce chemical energy (food) for the other animals in the ecosystem.
- 2 Dead plant material falls into the water and is broken down to form mud or detritus. Bacteria in the mud release dissolved nutrients into the water.
- 3 Worms, crabs and sandhoppers rely directly on detritus for food.
- 4 Phytoplankton is the food source for oysters.
- 5 The first-order consumers are worms, crabs, sandhoppers, molluscs and oysters.
- 6 Prawns, smaller fish, crabs, sandhoppers and molluscs are eaten by birds.
- 7 Both prawns and smaller fish rely on worms for food.
- 8 Canal developments will cause a rapid decline in the amount of mud and in the supply of plant material in the mud. This will decrease the food available for worms, crabs, sandhoppers, molluscs and oysters. As a result, the populations of all of these animals will decrease.
- 9 The population of oysters will decline as a result of canal development because their primary source of food, phytoplankton, rely on dissolved organic substances from the detritus to survive. As the supply of these nutrients declines due to loss of mangroves, seagrass and saltmarsh, the amount of phytoplankton will decline, causing a subsequent decline in the oyster population.
- 10 The population of birds would decline as a result of canal development because all of their sources of food would decline due to their direct or indirect reliance on plant material in detritus.

Activity 3: Environmental effects on the abundance of sand hoppers on sandy beaches



- Animals are most abundant in spring and early summer. They are least abundant in late summer to late winter.
- There is no obvious pattern. There are occasional irregular peaks.
- Water temperature is at its lowest in June/July and rises to higher temperatures from October to April/May.
- Months with a high wavepower preceded an increase in the number of animals. An increase in the number of animals preceded an increase in water temperature. A decrease in the number of animals began during the period with the highest water temperature.
- A month of high wavepower might have caused a subsequent increase in the number of animals after June 1988 and July 1989. However, this didn't occur after March 1990. High water temperatures (above 20°C) for about 4 months may cause a decrease in the number of animals.
- Abundance of predators, abundance of food, air or sand temperature, the type of sand (eg. particle size or how easy it is to burrow into), pollution and the amount of oxygen in the water could all cause a change in the number of animals.

Activity 4: Nest predation in Australian urban environments



Interpreting the data

- 1 Firstly, they are part of the largest category (large birds) that left prints in the plasticine eggs. Secondly, they were the most frequently observed nest predators in the volunteers' gardens. Thirdly, the rate of egg predation is higher in gardens where currawongs are more often seen.
- 2 Cat imprints rarely appeared on the plasticine eggs and they were rarely observed predating in the artificial nests.
- 3 Cats may only predate on nests which are often visited by adult birds. Therefore, this study does not give any real evidence of egg predation by cats.

Statement	definitely true	might be true	no evidence	might be false	definitely false
Big birds caused 64% of the imprints on the plasticine eggs.	✓				
13% of the predation observed by the volunteers was by currawongs.					✓
There is more predation of nests in gardens where currawongs are often observed.	✓				
Cats are major predators of small birds.		✓	✓	✓	
Currawongs eat more small bird eggs than any other predators.		✓			
Predation of small birds by currawongs is greater than predation by cats.		✓	✓	✓	
Currawongs are a significant nest predator in urban environments.	✓				

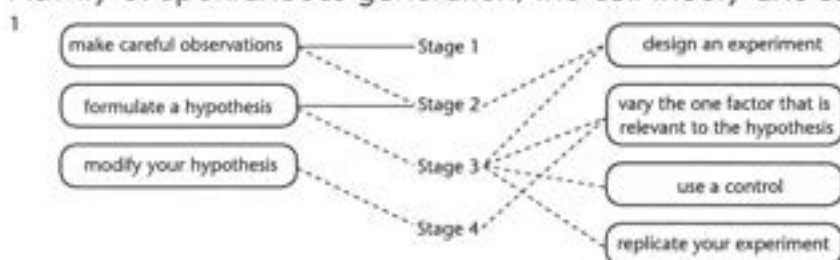
Recommendations

Recommendation	Evidence from the study that supports the recommendation
Plant more dense shrubs in gardens to provide small birds with more protection from predators.	The number of small birds is probably reduced by predation.
Reduce the number of berry-producing plants in gardens. These plants attract large birds like currawongs because they provide them with food.	Large birds such as currawongs are significant predators of small birds.
Do not encourage large birds into gardens by feeding them.	Large birds such as currawongs are significant predators of small birds.

Further studies

- 5 Firstly, artificial nests and eggs were used and predators might behave quite differently with real nests. Secondly, factors other than nest predation may be responsible for reducing the small bird population.
- 6 Conduct studies on real nests and investigate other factors affecting small bird populations.

Activity 5: Spontaneous generation, the cell theory and scientific process



- 2

Stage 1 — Worms appear in dead animals, devour the meat and then disappear.

Stage 2 — Worms are immature flies. Worms form a hard shell then hatch into flies.

Stage 3 — The worms grew from something left on the meat by flies OR worms can only grow when supplied with fresh air.

Stage 4 — The worms grew from something left on the meat by flies.

- 3 He showed that the maggots that appeared on rotting meat had not been generated spontaneously, but had grown from eggs laid by two species of flies.
- 4 The belief in spontaneous generation had remained unchallenged for centuries. People believed it to be true and so were unwilling to accept the alternative explanation that life comes from pre-existing life. The belief in spontaneous generation had been explained in religious terms, so challenging it was like challenging the authority of the church.

Activity 6: Size of cells and cell organelles

What to do

Plant cells	size (µm)	animal cells	size (µm)	other cells	size (µm)	plant cell organelles	size(µm)
phloem	200 long, 40 wide	cheek	50	bacterial	1-3	nucleus	10
pallisade	50 long, 20 wide	skin	15-20	amoeba	600	vacuole	10
epidermal	20 long, 10 wide	red blood	5-10	paramecium	300	chloroplasts	3-5

Interpreting the data

- Phloem cells from plant stems.
- Bacterial cells.
- Plant cells vary in size from 20 µm to 200 µm.
- The smallest animal cells are red blood cells which are 5 µm and the biggest animal cells are cheek cells which are 50 µm.
- Bacterial cells are 10-50x smaller than animal cells.

Classification stage

All living things are made of tiny parts called cells. All cells are very small but some cells are much smaller than others. Even though cells are very small, they contain organelles which are even smaller.

Description stage

Plant cells are larger than the cells of other types of organisms. They range in size from 20 µm for an epidermal cell to 200 µm for a phloem cell. Animal cells are generally smaller than plant cells. For example, red blood cells are about 5 µm. Skin cells are about 10 µm wide and 20 µm long. The largest animal cells are cheek cells which are about 50 µm. Bacterial cells are the smallest (1-3 µm) and protozoans are the biggest (up to 600 µm). Cell organelles also vary in size. In plant cells, the biggest organelles are the vacuoles which may fill most of the cell. By contrast, the chloroplasts are only about 5 µm in diameter.

Questions requiring research or interpretation

- Cells are microscopic because if they were any bigger, the demand for materials would be greater than the ability of the cell membrane to absorb them. The size of cells is limited by the SA/V (surface area to volume ratio). As cells get larger, their SA/V decreases. This means that the demand for materials such as oxygen, water, minerals and food (which is determined by the volume of the cell) increases faster than the surface area through which these materials move into the cell. Larger multicellular organisms overcome this problem by having specialised cells and organ systems (respiratory and circulatory) to supply their cells.
- Although bacterial cells are generally smaller than other types of cells, they are many times larger than viruses. In fact, many viruses invade bacterial cells.

Activity 7: Light and photosynthesis

Analysing the experiment

- Spirogyra* was used because in most plant cells, the chloroplasts are much smaller and less visible than in *Spirogyra*.
- Aerobic bacteria grow best where there is plenty of oxygen. Since photosynthesis produces oxygen, the bacteria should therefore grow best where photosynthesis is taking place.
- To find out which parts of the *Spirogyra* undertake photosynthesis and to determine what type of light is required.

Observations

- White light shining on chloroplasts.
- Red light shining on chloroplasts.
- The bacteria grew most rapidly next to the chloroplasts.
- The bacteria grew most rapidly when it was close to chloroplasts that were exposed to light.

Making links

- Photosynthesis occurs in chloroplasts that are exposed to light.
- Photosynthesis occurs when chloroplasts are exposed to red light but not green light.
- Treatment 2 is like a control treatment. It shows that every part of the chloroplast can photosynthesise. Without it you wouldn't be sure whether the effects in Treatment 1 and 3 were due to light or due to other factors.

Conclusions

- To find out if photosynthesis occurs in chloroplasts and if photosynthesis occurs in different coloured light.
- Photosynthesis occurs in chloroplasts that are exposed to white light. Photosynthesis occurs in white light but not in green light.



- The differences are in how much carbon dioxide and water were used, the fact that glucose is produced, and the fact that the chemical energy produced is in the form of ATP.
- Add another control treatment in purely dark conditions to show that the growth in treatments 1 and 2 is due to light.
- Green pigments reflect green light and absorb other colours. Therefore, it must be the green pigments in chloroplasts that carry out photosynthesis.

Activity 8: Using radioactive carbon to trace the path of carbon in plants.

Describing the data

- 1 Yes
- 2 It went both up and down the stem.
- 3 Cutting the phloem prevented the movement of the ^{14}C .
- 4 The tip or stem apex of the plant

Making links

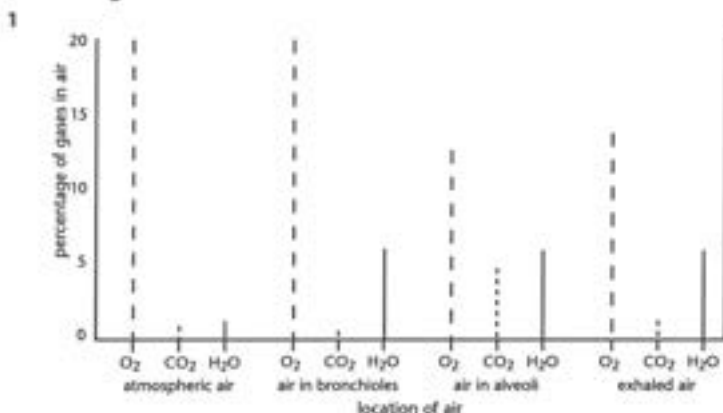
- 5 Both up and down.
- 6 It moved down into the roots but it only moved up until it reached the place where the phloem was cut.
- 7 It moved up into the stem apex but it only moved down until it reached the place where the phloem was cut.
- 8 It moved up and down the stem but only until it reached the places (above and below the leaf) where the phloem had been cut.

Interpretation

- 9 The phloem had been cut above the leaf in which ^{14}C was supplied. This prevented the movement of ^{14}C to the stem apex.
- 10 Because the phloem had been cut below the leaf in which ^{14}C was supplied. This prevented the movement of ^{14}C to the roots.
- 11 Yes, cutting the phloem prevented movement of carbon in the plant.
- 12 Glucose, because it is the product of photosynthesis that contains carbon.
- 13 After glucose is formed by photosynthesis, it moves both up and down the plant.

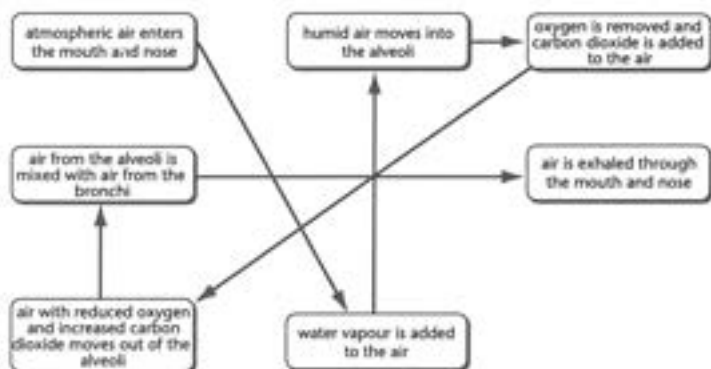
Activity 9: What happens to the air that we breathe in?

Processing the data



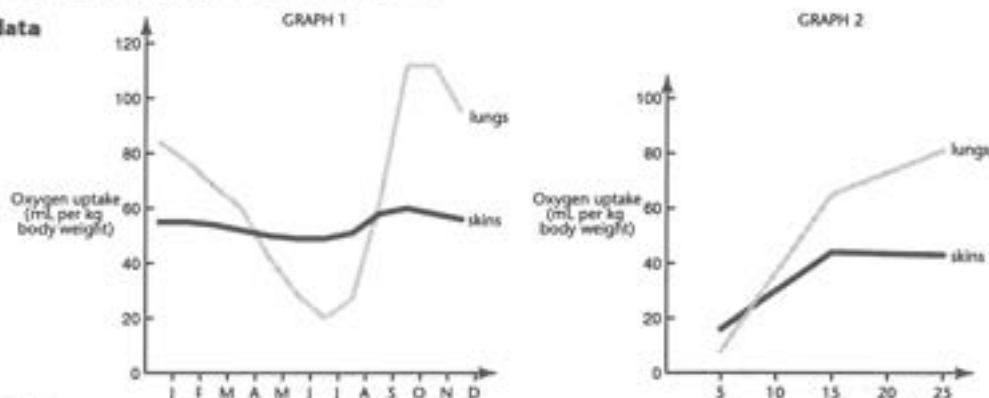
Interpreting the data

- 2 The percentage of water vapour in the air increases to about 6% soon after it enters the body through the mouth and nose and remains the same until after it is breathed out.
- 3 Moisture in the mucous linings of the nose, trachea and bronchi evaporates to moisten the air. Also, air is heated so it is able to hold more water vapour.
- 4 Helps prevent water loss in the alveoli, where the surface area is very high and therefore the risk of water loss is high.
- 5 Oxygen percentage in the air is reduced while the air is in the alveoli.
- 6 Oxygen moves from the air, through the walls of the alveoli and into the blood which carries it back to the heart.
- 7 The oxygen is transported to the body cells for use in respiration.
- 8 The percentage of carbon dioxide in the air increases while it is in the alveoli.
- 9 The carbon dioxide diffuses out of the blood, through the walls of the alveoli and into the air.
- 10 The carbon dioxide, which is a waste product of respiration in the cells, is removed from the body.
- 11



Activity 10: Gas exchange in amphibians

Graphing the data



Describing the data

- Oxygen uptake by the skin remains relatively constant throughout the year with a slight decrease in winter.
- January, February, March, April, September, October, November, December (spring, summer and autumn).
- June, July, August.
- a Summer.
b Winter.
- Skin.
- Lungs.

Making links

- Summer. Body temperature is higher so respiratory rate will be higher since all chemical reactions (including respiration) happen faster at higher temperatures.
- Because respiratory rate is higher in summer.
- Because skin remains the same size all year round.
- By breathing more often.
- Endotherms have a higher respiratory rate in winter because more energy is required to maintain their body temperature. Ectotherms' respiratory rate is lower in winter because their body temperature is lower. All chemical reactions slow down at lower temperatures.

Activity 11: Comparing the digestive systems of different mammals

Making measurements and tabulating data

Animal	total length of digestive system (cm)	length of body (cm)	length of digestive system divided by length of body	Does the animal have enlarged organs for microbial fermentation?
koala	500	51	10	very large elongated caecum
human	625	180	3.6	no
sheep	3200	110	30	large complex stomach and enlarged caecum

Making links

- No, the length of the digestive system varies greatly in the three animals. This is shown by the data about the length of the digestive system divided by the body length. Sheep have much longer digestive systems than humans despite having a shorter body length.
- Removing the effect of the length of the body makes it possible to compare the relative lengths of the digestive systems.
- Yes, the animals eating only fibrous plant material (sheep and Koalas) have much longer digestive systems than humans, who eat a bigger variety of foods.
- Humans. The diet of humans includes a wide variety of plant and animal products, many of which are easy to digest.
- Sheep. The diet of sheep is grass and other pasture plants, many of which are very fibrous and difficult to digest.
- Both animals have relatively long digestive systems but the digestive system of the sheep is longer. Both have large to very large caecums. However, the Koalas' caecum is very large in diameter compared to the sheep.
- Both sheep and Koalas eat highly fibrous plant material. If the length of the digestive system was the most important requirement for highly fibrous diets, they would both have equally long digestive systems.
- The koala has an extremely large and fat caecum in which to ferment eucalyptus leaves.

Making conclusions

- Animals that eat hard or fibrous plant material need a long digestive system with large organs in which this microbiological fermentation can take place. By contrast, mammals that have a diet that is easier to digest have shorter and simpler digestive systems without fermentation chambers.

Activity 12: Chemical evolution and the origin of life

Describing the data

From evidence box 1:

- Hydrogen (H_2), methane (CH_4), ammonia (NH_3), nitrogen (N_2), hydrogen sulfide (H_2S) and water (H_2O) were present in the atmosphere and elements (such as sulfur) and minerals were present in the sea.
- High levels of lightning, ultra violet radiation and radioactive decay.

From evidence box 2:

- 3 Water, ammonia, methane, hydrogen.
- 4 Heat and electrical spark discharge.
- 5 Amino acids and other simple organic compounds were formed.
- 6 It is possible that the conditions present on Earth could have led to the evolution of organic chemicals which became the building blocks of life.

From evidence box 3:

- 7 Points 3 and 4. Oparin's and Fox's work showed that proteins, nucleic acids and polysaccharides will form under the right conditions and that membrane-like structures will also form.

From evidence boxes 4 & 5:

- 8 Many species of modern bacteria are capable of utilising the materials present on Earth before life began. Also, there is fossil evidence of similar bacteria that lived early in the history of life on Earth.

Interpretation

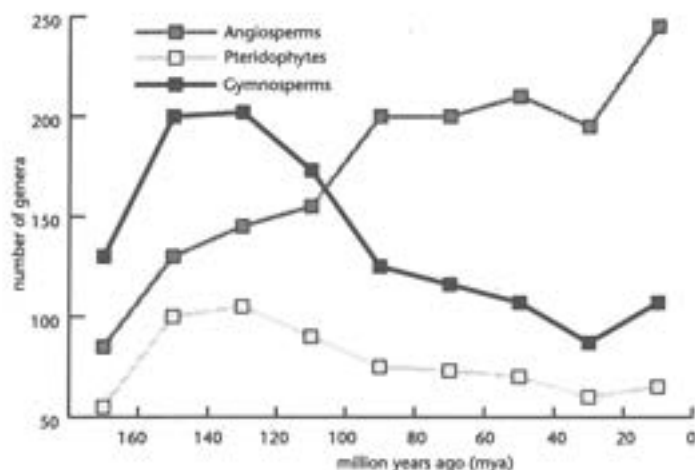
- 1 Firstly, organic molecules may have arisen before life. Secondly, the chemical evolution of the chemical building blocks of life is possible. Thirdly, early bacteria had a role in the beginnings of life on Earth.
- 2 Most components are backed by at least some evidence, although there is little evidence presented for the spontaneous development of complex membranes.

Activity 13: The role of prokaryotes in the evolution of life**Interpreting the data**

- 1 The first organic molecules, such as amino acids, ATP, nucleic acids and protein had appeared.
- 2 Chemosynthesis and anaerobic respiration. Other pathways required oxygen which was not present in the atmosphere.
- 3 It produced large quantities of oxygen and resulted in chemical changes in the ocean, atmosphere and rocks, and the formation of the ozone layer.
- 4 Aerobic respiration is only possible in the presence of oxygen. Photosynthetic cyanobacteria produced oxygen.
- 5 Eukaryotes needed prokaryotes to use inside their cells as mitochondria and chloroplasts.
- 6 Some prokaryotes have remained in relatively stable environments and so have remained relatively unchanged.
- 7 Chemical evolution → chemosynthetic bacteria → photosynthetic cyanobacteria → oxygen accumulation and ozone layer formation → aerobic respiration → cytochromes → eukaryotic cells incorporated prokaryotic cells → multicellular plants evolved from unicellular algae.

Making a concept map

Each pair of students should make their own. There is no one correct result!

Activity 14: Classification and diversity of plant groups over the past 160 million years**Graphing the data****Interpreting the data**

- 1 About 2000 million years ago.
- 2 About 450 million years ago.
- 3 Pteridophytes and Gymnosperms.
- 4 Angiosperms.
- 5 Gymnosperms and Pteridophytes increased until about 120 million years ago and have been basically declining ever since. Angiosperms have been increasing and now dominate the Gymnosperms and Pteridophytes.
- 6 Gymnosperms.
- 7 Angiosperms.
- 8 To live on land, plants need to be able to obtain, transport and conserve water. To do this they need true roots, stems, leaves, xylem and phloem. Enclosed seeds enable Angiosperms to live in a wider range of habitats.
- 9 The ability to reproduce in drier habitats. Compared to Pteridophytes, they have different reproductive adaptations that have enabled them to live in these drier habitats.
- 10 The reproductive adaptation of enclosed seeds has resulted in Angiosperms' greater diversity and their ability to grow in most habitats.
- 11 They may have been unsuccessfully competing with the Angiosperms, which have increased over the same period of time.

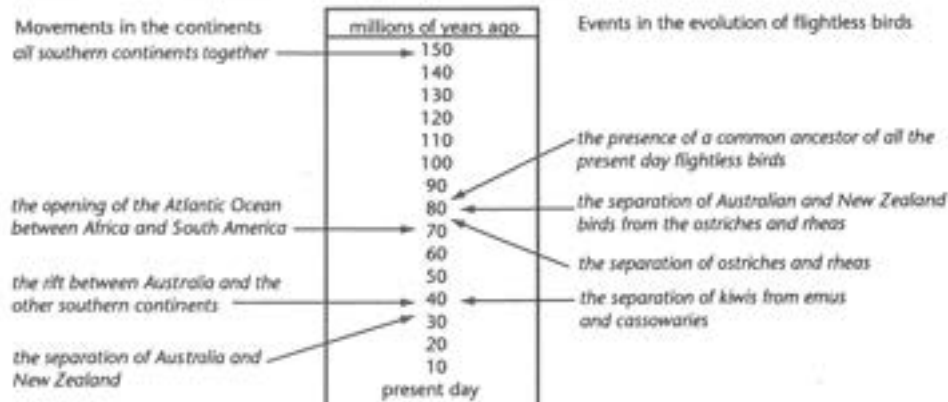
Activity 15: Continental drift and the distribution and evolution of flightless birds

Describing the data

1

Event or evidence	When did it occur? (millions of years ago)
all southern continents were together	150
the rift between Australia and the other southern continents	39
the opening of the Atlantic Ocean between Africa and South America	65
the separation of Australia and New Zealand (not on the diagrams)	40
the presence of a common ancestor of all the present day flightless birds	90
the separation of Australian and New Zealand birds from the ostriches and rheas	80
the separation of ostriches and rheas	80
the separation of kiwis from emus and cassowaries	40

2



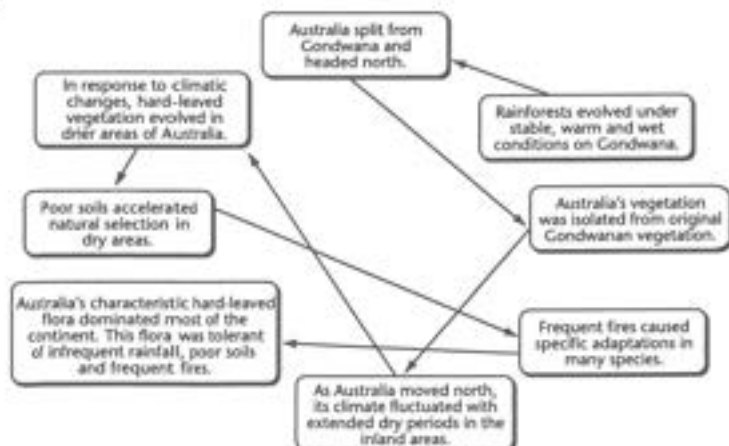
Making links

- All of the birds have similar DNA. The bigger the difference between 2 birds, the older their common ancestor is.
- It supports it, because all birds come from continents that used to be joined.
- Yes, the 2 continents began separating between 115 and 65 million years ago and DNA evidence suggests ostriches and rheas shared a common ancestor about 80 million years ago.
- Yes, kiwis and Australian birds shared a common ancestor about 45 million years ago and the 2 countries separated 40 million years ago.

Activity 16: The evolution of Australian plants

Questions

- Broad-leaved rainforest
- Rainforests of Gondwana
- Antarctica and South America
- Warm and wet conditions
- About 35 million years ago, when Australia split from Gondwana
- Cooler, more changeable climate with dry periods
- Rainforest contracted to areas around the coast which remained warm and wet. New species evolved in areas where climatic fluctuations occurred, such as species tolerant of dry periods.
- Both Australian and South American rainforests come from the same original forests and have experienced similar environmental conditions since.
- The vegetation in both continents has evolved for more than 35 million years under different environmental conditions. For example, Australia's environment is characterised by a variable climate, very poor soils and frequent bushfires.
-



Activity 17: The evolution of Australian mammals

Describing the data

- 1 These mammals were tiny and mouse-sized.
- 2 Marsupials were the most common in South America which was near Australia.
- 3 No, apart from South America, most other continents had mostly placental mammals whereas Australia had marsupials and monotremes.
- 4 Australia became drier, the forests receded and the grasslands spread.
- 5 Australia made contact with south-eastern Asia and the lush forests declined and grasslands spread.
- 6 Mammals became very large and were called the megafauna. Australia became cooler and drier, the rainforests receded and the grasslands spread.
- 7 Dingos and people arrived and climatic conditions fluctuated.

Making links

- 8 Modern Australian mammals may be predominantly marsupials and monotremes because the mammals present in Australia before it split from Gondwana were also marsupials and monotremes. The mammals in other parts of the world were, and still are, predominantly placentals.
- 9 The increasing dry periods and changing vegetation resulted in diverse habitats that may have resulted in more diverse fauna.
- 10 Rodents may have migrated from south-eastern Asia because Australia contacted south-eastern Asia during the Miocene Epoch. Also, rodents may have evolved from Australian mammals, although this is unlikely.
- 11 The appearance of the megafauna may have resulted from the vast areas of suitable habitat that was due to the expansion of grasslands and eucalypts.
- 12 The megafauna may have become extinct because of: predation by dingos, environmental change, predation by humans, fluctuating climatic conditions or a combination of all of these.

Conclusions

- 13 Australian mammal fauna is dominated by marsupials and monotremes because of a combination of the following factors:
 - △ marsupials and monotremes dominated the original population of mammals in Australia
 - △ isolation of Australia from the other continents due to continental drift
 - △ changes in environmental conditions leading to natural selection and speciation.

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